

**Design Requirements**

Cooling Capacity **48,000 BTU/h**  
 Configuration **Downflow**

- R-32 chlorine-free refrigerant
- High-efficiency Copeland scroll compressor(s)
- Copper tube / aluminum fin coils
- High-capacity, steel-cased filter dryer
- High- and Low-pressure switches
- Lugged electrical contactor
- Single-point power
- Color Coded Wiring
- Foil Faced Internal Insulation
- Heavy-gauge, galvanized-steel cabinet
- UV-resistant powder-paint finish
- Full perimeter base rail

**Unit Electrical Data**

Power Supply **460/3/60 VOLT/PH**  
 Total Unit MCA **11.9 A**  
 Total Unit MOP **15.0 A**  
 Indoor Motor HP **1.2 HP**  
 Total Unit FLA **10.30 A**  
 • Compressor RLA **6.2**  
 • Blower FLA **2.5**  
 • Outdoor Fan FLA **1.6**  
 SCCR **5 kA**

**Model Number: DHH0484D000031C**

**Code String: DHH0484DXXXCBDCXFEXXX**

**Tag: 4-Ton - HE - 460/3 - 0kW**

**Cooling Performance \*4**

Total Capacity **48,225 BTU/h**  
 Sensible Capacity **38,116 BTU/h**  
 Cooling Stages **Two**  
 Refrig Type **R-32**  
 Compressor Type **Copeland Scroll**  
 Pressure Switches **High & Low Pressure (std)**  
 Outdoor Design Temp **95.00°F**  
 Entering Air Temp(db) **80.00°F**  
 Entering Air Temp(wb) **67.00°F**  
 Leaving Air Temp(db) **58.30°F**  
 Leaving Air Temp(wb) **57.50°F**  
 Efficiency (M1) **16.8 SEER2/13.0 EER2**

**Heating Performance \*4**

Heat Type **Heat Pump**  
 LAT (47°F) **81.60**  
 BTU/h (47°F) **46000 @ 70°F Indoor DB**  
 HSPF2 **7.7**

**Supply Air Blower Performance**

ACFM **1600 cfm**  
 SCFM **1600 cfm**  
 ESP **0.50 IWG**  
 Ultra Low-Leak Downflow Economizer For DDC Controls W/ Enthalpy Sensor **0.09 IWG**  
 Total ESP **0.59 IWG**  
 Blower Type **Direct Drive - Standard Static**  
 Elevation **0 ft**

**Physical Attributes**

Base Unit Oper. Weight (lbs) **621**  
 Filter(s) **14"X20"X2" Qty: 4**

**Outdoor sound (db) at 60 Hz**

| Model | A-Weighted | 63  | 125  | 250  | 500  | 1000 | 2000 | 4000 | 8000 |
|-------|------------|-----|------|------|------|------|------|------|------|
| 048   | 77.0       | N/A | 80.6 | 73.0 | 73.5 | 74.3 | 67.4 | 62.0 | 53.0 |

db - decibel

**Notes:**

1 Outdoor sound data is measured in accordance with AHRI standard 270.

2 Measurements are expressed in terms of sound power. Do not compare these values to sound pressure values because sound pressure depends on specific environment factors which normally do not match individual applications. Sound power values are independent of the environment and therefore more accurate.

3 A-weighted sound ratings filter out high and very low frequencies, to better approximate the response of average human ear. A-weighted measurements for Daikin units are taken in accordance with AHRI standard 270.

4 All air temps are measured at the unit's inlet and outlet. Air temps are not measured at the coil.

## LIST OF ACCESSORIES

Model Number: DHH0484D000031C; Code String: DHH0484DXXXCBDCXFEXXX; Tag: 4-Ton - HE - 460/3 - 0kW;

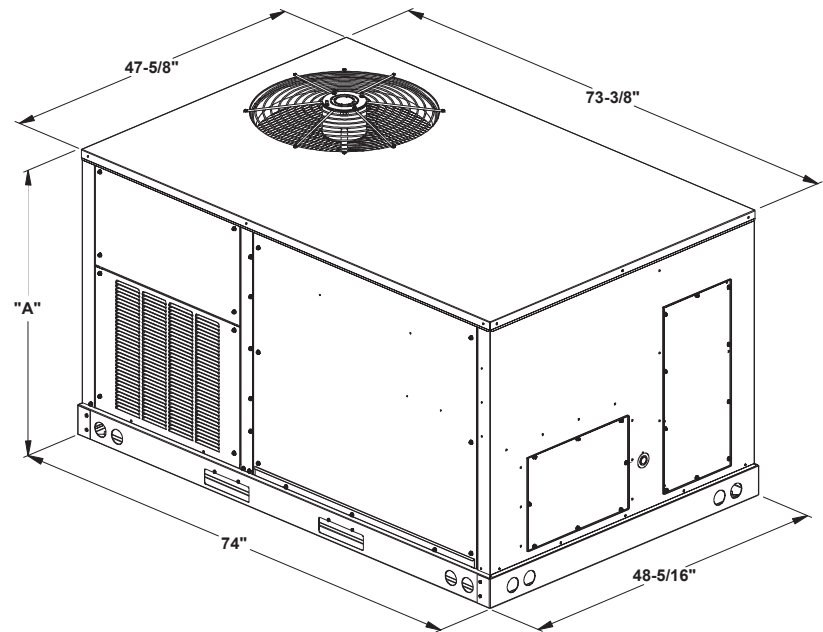
| Accessories |          |                 |                                                                           |            |     |
|-------------|----------|-----------------|---------------------------------------------------------------------------|------------|-----|
| Fld Inst    | Fct Inst | Item #          | Description                                                               | Dimensions | Lbs |
|             | Y        | HGDRSM02FI      | Hinged Panels                                                             |            | 34  |
|             | Y        | DSC060A54YH01FI | Non Fused Disconnect Switch                                               | Fact Inst  | 5   |
|             | Y        | HAILGD48060HEFI | Hail Guard                                                                |            | N/A |
|             | Y        | TTBCKHE01FI     | Thru-The-Base Connections                                                 |            | 1   |
| Y           |          | 0130L00223      | Outdoor Temperature                                                       |            | N/A |
| Y           |          | 0221L00015      | 24" Roof Curb                                                             |            | 109 |
| Y           |          | 0270L02827      | Ultra Low-Leak Downflow Economizer For DDC Controls W/<br>Enthalpy Sensor |            | N/A |
| Y           |          | 0130L00226      | DDC Space Dry-Bulb Sensor Wall Mounted                                    |            | N/A |
| Y           |          | 0130L00224      | DDC Space Enthalpy Sensor Wall Mounted                                    |            | N/A |
|             | Y        | CONVOUTNP036072 | Non Powered GFCI Convenience Outlet                                       | Fact Inst  | 2   |
|             | Y        |                 | DDC Communicating Controller                                              | 8.5 × 6.05 | 2   |
|             |          |                 | Base Unit Oper. Weight                                                    |            | 621 |
|             |          |                 | Total Weight                                                              |            | 774 |

CORNER WEIGHTS DIMENSIONS

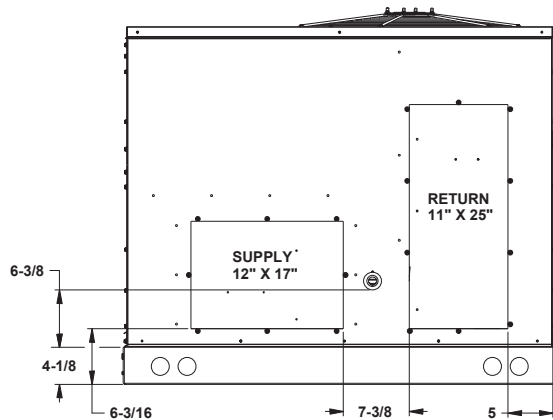


CORNER AND CENTER OF GRAVITY LOCATIONS

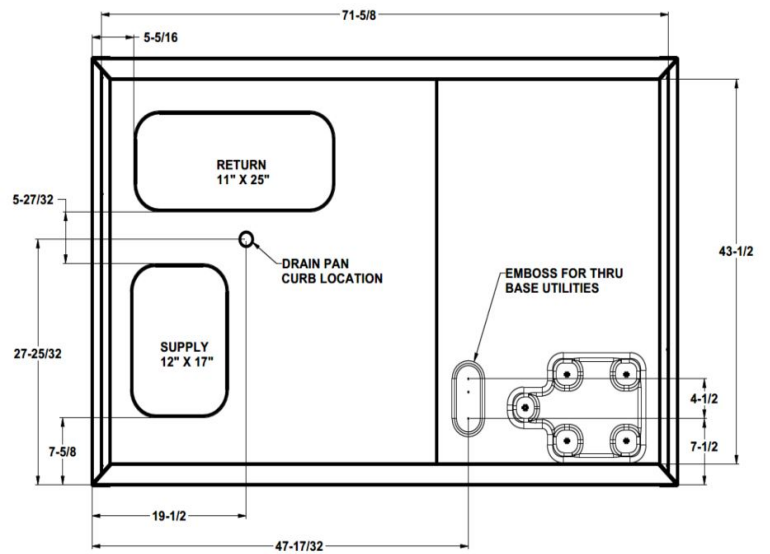
| Model    | Shipping Weight (lbs) | Operating Weight (lbs) | Corner Weights (lbs) |     |     |     | Length | Width  |
|----------|-----------------------|------------------------|----------------------|-----|-----|-----|--------|--------|
|          |                       |                        | A                    | B   | C   | D   | X (in) | Y (in) |
| DHH0484D | 679                   | 621                    | 166                  | 176 | 112 | 167 | 33.30  | 26.70  |



| MODEL SIZE | DIM "A" |
|------------|---------|
| 3 ton HP   | 43-1/2" |
| 4 ton HP   | 43-1/2" |
| 5 ton HP   | 43-1/2" |
| 6 ton HP   | 53-3/4" |



HORIZONTAL DISCHARGE



BOTTOM VIEW OF UNIT  
VERTICAL DISCHARGE





# RH OUTSIDE AIR

## Relative Humidity, Outside Air, Thermistors

The ACI Relative Humidity with Thermistor Outside Air Series utilizes a thermoset polymer capacitive sensing element with factory applied hygroscopic filter to deliver a proportional analog current or voltage output signal. The hygroscopic filter provides added resistance to moisture, dust, and other chemicals for greater long term reliability. The RH Outside Air transmitter features integral DIP switches for field selection of the proper output signal and supply voltage to meet your applications requirements. Each unit also contains 0%, 50%, and 100% test options to verify that the transmitter is both working and wired properly. Field calibration also can be performed by using the increment and decrement calibration DIP switches without the need to replace the sensing element. These enhancements provide increased flexibility and outstanding long-term reliability. Outside Air configurations feature a

weatherproof Euro style enclosure with gasketed cover and conformally coated circuit boards for added moisture and chemical resistance. Three point NIST Calibration Certificates are available.

**Applications:** Monitor Outdoor Air Humidity, Economizer Control, Psychrometric calculations such as Enthalpy and Dew point, Wash down Applications

**The ACI RH Thermistor Outside Air is covered by ACI's Five (5) Year Limited Warranty. The warranty can be found in the front of ACI's Sensors & Transmitters catalog, as well as on ACI's website, [www.workaci.com](http://www.workaci.com).**

### PRODUCT SPECIFICATIONS

|                                                                                        |                                                                                                                                                               |                                                    |
|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| <b>RH Supply Voltage</b>                                                               | <b>4-20 mA:</b> 250 Ohm Load: 15 - 40 VDC / 18 - 28 VAC   <b>500 Ohm Load:</b> 18 - 40 VDC / 18 - 28 VAC                                                      |                                                    |
| <b>(Reverse Polarity Protected):</b>                                                   | <b>0-5 VDC:</b> 12 - 40 VDC / 18 - 28 VAC   <b>0-10 VDC:</b> 18 - 40 VDC / 18 - 28 VAC                                                                        |                                                    |
| <b>RH Supply Current (VA):</b>                                                         | <b>Voltage Output:</b> 8 mA maximum (0.32 VA)   <b>Current Output:</b> 24 mA maximum (0.83 VA)                                                                |                                                    |
| <b>RH Output Load Resistance:</b>                                                      | <b>4-20 mA:</b> 700 Ohms maximum   <b>0-5 VDC or 0-10 VDC:</b> 4K Ohms Minimum                                                                                |                                                    |
| <b>RH Output Signal:</b>                                                               | <b>2-wire:</b> 4 - 20 mA (Factory Default)   <b>3-wire:</b> 0-5 or 0-10 VDC and 4 - 20 mA (Field Selectable)                                                  |                                                    |
| <b>RH Accuracy @ 77°F (25°C):</b>                                                      | +/- 1% over 20% RH Range between 20 to 90%   +/- 2%, 3%, or 5% from 10 to 95%                                                                                 |                                                    |
| <b>RH Measurement Range:</b>                                                           | 0-100%                                                                                                                                                        |                                                    |
| <b>Operating RH Range:</b>                                                             | 0 to 95% RH, non-condensing (Conformally Coated PCB's)                                                                                                        |                                                    |
| <b>Operating Temperature Range:</b>                                                    | -40 to 140°F (-40 to 60°C)                                                                                                                                    |                                                    |
| <b>Storage Temperature Range:</b>                                                      | -40 to 149°F (-40 to 65°C)                                                                                                                                    |                                                    |
| <b>RH Stability   Repeatability   Sensitivity:</b>                                     | Less than 2% drift / 5 years   0.5% RH   0.1% RH                                                                                                              |                                                    |
| <b>RH Response Time (T63):</b>                                                         | 20 Seconds Typical                                                                                                                                            |                                                    |
| <b>RH Sensor Type:</b>                                                                 | Capacitive with Hydrophobic Filter                                                                                                                            |                                                    |
| <b>RH Transmitter Stabilization Time:</b>                                              | 30 Minutes (Recommended time before doing accuracy verification)                                                                                              |                                                    |
| <b>RH Connections   Wire Size:</b>                                                     | Screw Terminal Blocks (Polarity Sensitive)   16 (1.31 mm <sup>2</sup> ) to 26 AWG (0.129 mm <sup>2</sup> )                                                    |                                                    |
| <b>RH Terminal Block Torque Rating:</b>                                                | 4.43 to 5.31 lb-in (0.5 to 0.6 Nm)                                                                                                                            |                                                    |
| <b>RH NIST Test Points:</b>                                                            | <b>Default Test Points:</b> 3 Points (20%, 50% & 80%)<br><b>1% NIST Test Points:</b> 5 Points within selected 20% Range (ie. 30%-50% are 30, 35, 40, 45 & 50) |                                                    |
| <b>Nominal Thermistor Resistive Output @ 77°F (25°C)</b>                               | <b>RHx-1.8K Series:</b> 1.8KΩ (Red/Yellow)                                                                                                                    | <b>RHx-10KS Series:</b> 10KΩ (White/Blue)          |
| <b>(Lead Wire Colors):</b>                                                             | <b>RHx-3K Series:</b> 3KΩ (White/Brown)                                                                                                                       | <b>RHx-10K-E1 Series:</b> 10KΩ (Gray/Orange)       |
|                                                                                        | <b>RHx-AN Series (Type III):</b> 10KΩ (White/White)                                                                                                           | <b>RHx-20K Series:</b> 20KΩ (Brown/Blue)           |
|                                                                                        | <b>RHx-AN-BC Series:</b> 5.238KΩ (White/Yellow)                                                                                                               | <b>RHx-50K Series:</b> 50KΩ nominal (Brown/Yellow) |
|                                                                                        | <b>RHx-CP Series (Type II):</b> 10KΩ (White/Green)                                                                                                            | <b>RHx-100KS Series:</b> 100KΩ (Black/Yellow)      |
|                                                                                        | <b>RHx-CSI Series:</b> 10KΩ (Green/Yellow)                                                                                                                    |                                                    |
| <b>Thermistor Accuracy 32-158°F (0-70°C):</b>                                          | +/- 0.36°F (0.2°C) except 10K-E1 Series: +/- 0.54°F (0.3°C)                                                                                                   |                                                    |
| <b>Thermistor Power Dissipation Constant:</b>                                          | 1.8K Series: +/- 0.9°F (0.5°C) @ 77°F (25°C) & +/- 1.8°F (1.0°C) from 32 to 158°F (0 to 70°C)                                                                 |                                                    |
| <b>Thermistor Sensor Response Time (T63):</b>                                          | 3 mW/°C except 1.8K Series: 1 mW/°C; 10K-E1 Series: 2 mW/°C                                                                                                   |                                                    |
| <b>Lead Wire Length   Conductor Size:</b>                                              | 10 Seconds nominal                                                                                                                                            |                                                    |
| <b>Insulation   Rating:</b>                                                            | 14" (35.6 cm)   22 AWG (0.65 mm)                                                                                                                              |                                                    |
| <b>Enclosure Specifications (Material, Flammability, Temperature, NEMA/IP Rating):</b> | Etched Teflon (PTFE) Colored Leads   Mil Spec 16878/4 Type E                                                                                                  |                                                    |
| <b>Sensing Tube Dimensions (Length x Diameter):</b>                                    | <b>"-EH" Enclosure:</b> ABS Plastic; UL94-V0; -40 to 140°F (-40 to 60°C)                                                                                      |                                                    |
|                                                                                        | <b>"-4X" Enclosure:</b> Polystyrene Plastic; UL94-V2; -40 to 158°F (-40 to 70°C); NEMA 4X (IP 66)                                                             |                                                    |
|                                                                                        | <b>"-EH" Models:</b> 3.00" (76.20 mm) x 1.125" (28.75 mm)                                                                                                     |                                                    |
|                                                                                        | <b>"-4X" Models:</b> 4.73" (120.14 mm) x 0.845" (21.46mm)                                                                                                     |                                                    |
| <b>Product Dimensions (L x W x D):</b>                                                 | See drawings on back of data sheet                                                                                                                            |                                                    |
| <b>Product Weight:</b>                                                                 | <b>A/RHx-xx-O Series:</b> 0.59 lbs. (0.27 kg)   <b>A/RHx-xx-O-4X Series:</b> 0.45 lbs. (0.204 kg)                                                             |                                                    |
| <b>Agency Approvals:</b>                                                               | CE, RoHS2, WEEE                                                                                                                                               |                                                    |



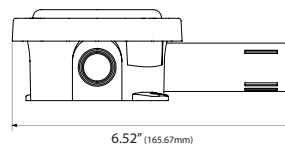
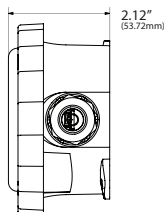
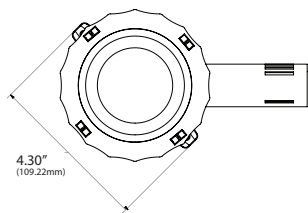
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HUMIDITY | ##

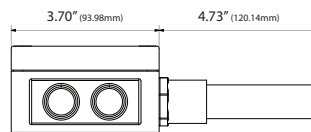
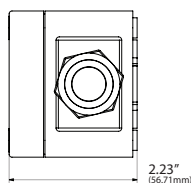
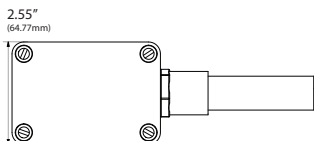


## DIMENSIONAL DRAWING

### Euro Enclosure [EH]



### NEMA 4X Enclosure [4X]



Front View

Right View

Top View

## CUSTOM ORDERING

Model # Example:

A/ RH2 CP O NIST

## MODEL #

| A. Sensor Series No Selection Required    | A/                                                                                                | A/ |
|-------------------------------------------|---------------------------------------------------------------------------------------------------|----|
| B. Accuracy Select One (1)                | RH1 = +/-1% (Specify a 20% Range between 20 to 90% RH)<br>RH2 = +/-2%   RH3 = +/-3%   RH5 = +/-5% |    |
| C. Temperature Sensor Select One (1)      | 1.8K   3K   10KS   AN (Type III)   AN-BC   CP (Type II)   CSI   10K-E1   20K   50K   100KS        |    |
| D. Configuration Select One (1)           | O = Outside Air (Euro Enclosure)   O-4X = Outside Air (NEMA 4X Enclosure)                         |    |
| E. Output Signal Select One (1)           | ---- = 4 to 20 mA (Default)   0 to 10 VDC (Field Selectable)   0 to 5 VDC (Field Selectable)      |    |
| F. NIST (Temperature & RH) Select One (1) | ---- = No NIST Certificate   NIST = NIST Certificate (3 Points)                                   |    |

**Note:** Outputs are field selectable between 4-20 mA, 0-5 VDC & 0-10 VDC





## RH ROOM

### Relative Humidity Room, Thermistor

The ACI Relative Humidity with Thermistor Room Series utilizes a thermoset polymer capacitive sensing element with a factory fitted hydrophobic filter to improve its moisture resistance. The sensing elements multilayer construction also provides excellent resistance in applications where dust, dirt, oils and common environmental chemicals are found. The RH room sensors include on board DIP switches which allow the user to select the desired output signal and can be powered by AC or DC power sources. Single point field calibration can be performed by using the increment and decrement calibration DIP switches to adjust your curve up or down in +/- 0.5% increments with each toggle of the corresponding switches. These enhancements provide increased flexibility and outstanding long-term reliability without the need to replace the sensors in the field. There are two enclosure options in this series which should satisfy most

commercial decors. Both enclosures feature four-way airflow to minimize self-heating. Three point NIST Calibration Certificates are available.

**Applications:** Humidification, Dehumidification, Monitoring Indoor Space Humidity, Clean Rooms, Hospitals, Process Control, Laboratories, Museums, Schools, Office Buildings, Data Centers, ESD (Anti-Static) Control

**The ACI RH Thermistors Room is covered by ACI's Five (5) Year Limited Warranty. The warranty can be found in the front of ACI's Sensors & Transmitters catalog, as well as on ACI's website, [www.workaci.com](http://www.workaci.com).**

### PRODUCT SPECIFICATIONS

|                                                                                                                                |                                                                                                                                                                     |                                                    |
|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| <b>RH Supply Voltage:</b>                                                                                                      | <b>4-20 mA:</b> 250 Ohm Load: 15 - 40 VDC / 18 - 28 VAC   <b>500 Ohm Load:</b> 18 - 40 VDC / 18 - 28 VAC                                                            |                                                    |
| <b>(Reverse Polarity Protected):</b>                                                                                           | <b>0-5 VDC:</b> 12 - 40 VDC / 18 - 28 VAC   <b>0-10 VDC:</b> 18 - 40 VDC / 18 - 28 VAC                                                                              |                                                    |
| <b>RH Supply Current (VA):</b>                                                                                                 | <b>Voltage Output:</b> 8 mA maximum (0.32 VA)   <b>Current Output:</b> 24 mA maximum (0.83 VA)                                                                      |                                                    |
| <b>RH Output Load Resistance:</b>                                                                                              | <b>4-20 mA:</b> 700 Ohms maximum   <b>0-5 VDC or 0-10 VDC:</b> 4K Ohms Minimum                                                                                      |                                                    |
| <b>RH Output Signal:</b>                                                                                                       | <b>2-wire:</b> 4 - 20 mA (Factory Default)   <b>3-wire:</b> 0-5 or 0-10 VDC & 4 - 20 mA (Field Selectable)                                                          |                                                    |
| <b>RH Accuracy @ 77°F (25°C):</b>                                                                                              | +/- 1% over 20% RH Range between 20 to 90%   +/- 2%, 3%, or 5% from 10 to 95%                                                                                       |                                                    |
| <b>RH Measurement Range:</b>                                                                                                   | 0-100%                                                                                                                                                              |                                                    |
| <b>Operating RH Range:</b>                                                                                                     | 0 to 95% RH, non-condensing                                                                                                                                         |                                                    |
| <b>Operating Temperature Range:</b>                                                                                            | 35 to 122°F (1.5 to 60°C)                                                                                                                                           |                                                    |
| <b>Storage Temperature Range:</b>                                                                                              | -40 to 149°F (-40 to 65°C)                                                                                                                                          |                                                    |
| <b>RH Stability   Repeatability   Sensitivity:</b>                                                                             | Less than 2% drift / 5 years   0.5% RH   0.1% RH                                                                                                                    |                                                    |
| <b>RH Response Time (T63):</b>                                                                                                 | 20 Seconds Typical                                                                                                                                                  |                                                    |
| <b>RH Sensor Type:</b>                                                                                                         | Capacitive with Hydrophobic Filter                                                                                                                                  |                                                    |
| <b>RH Transmitter Stabilization Time:</b>                                                                                      | 30 Minutes (Recommended time before doing accuracy verification)                                                                                                    |                                                    |
| <b>RH Connections   Wire Size:</b>                                                                                             | Screw Terminal Blocks (Polarity Sensitive)   16 (1.31 mm <sup>2</sup> ) to 26 AWG (0.129 mm <sup>2</sup> )                                                          |                                                    |
| <b>RH Terminal Block Torque Rating:</b>                                                                                        | 4.43 to 5.31 lb-in (0.5 to 0.6 Nm)                                                                                                                                  |                                                    |
| <b>RH NIST Test Points:</b>                                                                                                    | <b>Default Test Points:</b> 3 Points (20%, 50% & 80%)<br><b>1% NIST Test Points:</b> 5 Points within selected 20% Range (ie. 30%-50% are 30, 35, 40, 45 & 50)       |                                                    |
| <b>Nominal Thermistor Resistive Output @ 77°F (25°C) (Lead Wire Colors) Non-Linear NTC (Negative Temperature Coefficient):</b> | <b>RHx-1.8K Series:</b> 1.8KΩ (Red/Yellow)                                                                                                                          | <b>RHx-10KS Series:</b> 10KΩ (White/Blue)          |
|                                                                                                                                | <b>RHx-3K Series:</b> 3KΩ (White/Brown)                                                                                                                             | <b>RHx-10K-E1 Series:</b> 10KΩ (Gray/Orange)       |
|                                                                                                                                | <b>RHx-AN Series (Type III):</b> 10KΩ (White/White)                                                                                                                 | <b>RHx-20K Series:</b> 20KΩ (Brown/Blue)           |
|                                                                                                                                | <b>RHx-AN-BC Series:</b> 5.238KΩ (White/Yellow)                                                                                                                     | <b>RHx-50K Series:</b> 50KΩ nominal (Brown/Yellow) |
|                                                                                                                                | <b>RHx-CP Series (Type II):</b> 10KΩ (White/Green)                                                                                                                  | <b>RHx-100KS Series:</b> 100KΩ (Black/Yellow)      |
| <b>Thermistor Accuracy 32-158°F (0-70°C):</b>                                                                                  | <b>RHx-CSI Series:</b> 10KΩ (Green/Yellow)                                                                                                                          |                                                    |
|                                                                                                                                | +/- 0.36°F (0.2°C) except 10K-E1 Series: +/- 0.54°F (0.3°C)<br><b>1.8K Series:</b> +/- 0.9°F (0.5°C) @ 77°F (25°C) & +/- 1.8°F (1.0°C) from 32 to 158°F (0 to 70°C) |                                                    |
| <b>Thermistor Power Dissipation Constant:</b>                                                                                  | 3 mW/°C except 1.8K Series: 1 mW/°C; 10K-E1 Series: 2 mW/°C                                                                                                         |                                                    |
| <b>Thermistor Sensor Response Time (T63):</b>                                                                                  | 10 Seconds nominal                                                                                                                                                  |                                                    |
| <b>Temperature Connections   Wire Size:</b>                                                                                    | Screw Terminal Blocks   16 (1.31 mm <sup>2</sup> ) to 26 AWG (0.129 mm <sup>2</sup> )                                                                               |                                                    |
| <b>Temperature Terminal Block Torque Rating:</b>                                                                               | 4.43 to 5.31 lb-in (0.5 to 0.6 Nm)                                                                                                                                  |                                                    |
| <b>Enclosure Material (Color):</b>                                                                                             | <b>"-R2" Enclosure:</b> ABS (White)   <b>"-R" Enclosure:</b> ABS (Beige)                                                                                            |                                                    |
| <b>Enclosure Flammability Rating:</b>                                                                                          | UL94-HB                                                                                                                                                             |                                                    |
| <b>Product Dimensions (L x W x D):</b>                                                                                         | <b>"-R2" Enclosure:</b> 4.50" (114.3 mm) x 2.75" (69.85 mm) x 1.12" (28.45 mm)                                                                                      |                                                    |
|                                                                                                                                | <b>"-R" Enclosure:</b> 4.50" (114.3 mm) x 2.75" (69.85 mm) x 1.12" (28.45 mm)                                                                                       |                                                    |
| <b>Product Weight:</b>                                                                                                         | <b>A/RHx-xx-R2 Series:</b> 0.17 lbs. (0.077 kg)   <b>A/RHx-xx-R Series:</b> 0.17 lbs. (0.077 kg)                                                                    |                                                    |
| <b>Agency Approvals:</b>                                                                                                       | CE, RoHS2, WEEE                                                                                                                                                     |                                                    |

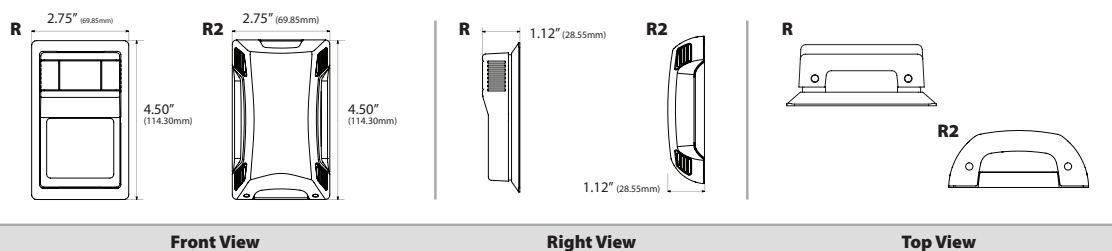


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HUMIDITY | ##



## DIMENSIONAL DRAWING



## CUSTOM ORDERING

Model # Example: A/ RH2 CP RSO NIST 0 Ohms 1K Ohms DA + to -

MODEL #

|                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                       |           |
|-------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>A. Sensor Series</b> No Selection Required                                                                           | A/                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>A/</b> |
| <b>B. Accuracy</b> Select One (1)                                                                                       | RH1 = +/-1% (Specify a 20% Range between 20 to 90% RH)<br>RH2 = +/-2%   RH3 = +/-3%   RH5 = +/-5%                                                                                                                                                                                                                                                                                                                     |           |
| <b>C. Temperature Sensor</b> Select One (1)                                                                             | 1.8K   3K   10KS   AN (Type III)   AN-BC   CP (Type II)   CSI   10K-E1   20K   50K   100KS                                                                                                                                                                                                                                                                                                                            |           |
| <b>D. Configuration</b> Select One (1)                                                                                  | R = Room<br>RO = Room with Override<br>RS = Room with Setpoint<br>RSO = Room with Setpoint and Override<br>R2 = Room<br>R2O = Room with Override<br>R2S = Room with Setpoint<br>R2SO = Room with Setpoint and Override                                                                                                                                                                                                |           |
| <b>E. Output Signal</b> Select One (1)                                                                                  | ---- = 4 to 20 mA (Default)   0 to 10 VDC (Field Selectable)   0 to 5 VDC (Field Selectable)                                                                                                                                                                                                                                                                                                                          |           |
| <b>F. NIST (Temperature &amp; RH)</b> Select One (1)                                                                    | ---- = No NIST Certificate   NIST = NIST Certificate (3 Points)                                                                                                                                                                                                                                                                                                                                                       |           |
| <b>Setpoint Configuration Options</b> Select Options below if RS, RSO, R2S or R2SO was selected as a Configuration (C). |                                                                                                                                                                                                                                                                                                                                                                                                                       |           |
| <b>1. Slidepots</b> Select One (1)                                                                                      | <b>Direct Acting</b> (Range in Ohms)<br>A01 = 0 to 100K   A02 = 0 to 20K   A03 = 0 to 10K   A06 = 4.75K to 24.75K<br>A07 = 10K to 30K   A08 = 1K to 11K   A09 = 0 to 2K   A10 = 0 to 1K<br>A11 = 2.05K to 3.05K   A12 = 0 to 400   A16 = 0 to 5K   A18 = 10K to 15K<br>A26 = 866 to 1,266   A29 = 7.87K to 27.8K<br><b>Reverse Acting</b> (Range in Ohms)<br>A04 = 1051.1 to 51.1   A14 = 10K to 0   A24 = 9.5K to 1K |           |
| <b>2. Setpoint Stickers</b> Select One (1)                                                                              | A3 = 18-28 DEG C   A4 = 20-30 DEG C   B4 = 55-85 DEG F   B7 = 60-90 DEG F   C5 = COOL/WARM<br>C6 = COOLER/WARMER   D3 = WARM/COOL   G5 = BLUE/RED (R2 Enclosure)                                                                                                                                                                                                                                                      |           |

Note: Outputs are field selectable between 4-20 mA, 0-5 VDC &amp; 0-10 VDC

## ACCESSORIES ORDERING

Model # Example: A/MOUNTING PLATE BEIGE R -OR- 106821

| Model #                   | Item # | Description                                             |
|---------------------------|--------|---------------------------------------------------------|
| A/MOUNTING PLATE BEIGE R  | 106821 | Wall Mounting Back Plate, Plastic, Beige ("R")          |
| A/MOUNTING PLATE WHITE R2 | 143369 | Wall Mounting Back Plate, Plastic, White ("R2")         |
| LOCKING COVER             | 107370 | Clear Thermostat Guard, Locking Cover, Low Profile      |
| A/ROOM-FOAM-PAD           | 125690 | 1/8" Foam Insulation Pad with Adhesive (3" x 2", Black) |





## ROOM

### Wall Mount Enclosures, Thermistor

The ACI Thermistor Room Series combines option flexibility with attractive styling in our “R2” or “R” style enclosures which include four-way air flow design too minimize self-heating to the sensor. These enclosures are offered in a White “R2” or Beige “R” color depending on the enclosure style. These units are designed to be mounted over a single gang junction box or hole in the wall using drywall anchors. Screw terminal blocks are available for making all connections to your building management system (network). An optional 1/8” Black foam pad with pressure sensitive adhesive is available to insulate the sensor from thermal drafts within the wall or wall surface. A 1/16” Hex driver is needed to secure the cover from being easily removed. The “LCD” option uses two temperature sensors to monitor the ambient air temperature in the space and is factory calibrated at a single point.

**Applications:** Space Temperature Sensing, Decorative Wall Sensor Applications, Office Buildings, Schools, Colleges, Commercial Buildings, OEM Opportunities

The ACI Thermistor Room is covered by ACI's Five (5) Year Limited Warranty. The warranty can be found in the front of ACI's Sensors & Transmitters catalog, as well as on ACI's website, [www.workaci.com](http://www.workaci.com).

### PRODUCT SPECIFICATIONS

|                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sensor Type   Sensor Curve:</b>                      | Thermistor   Non-Linear, NTC (Negative Temperature Coefficient)                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Number Temperature Sensing Points:</b>               | One                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Sensor Output @ 25°C (77°F):</b>                     | <b>A/1.8K Series:</b> 1.8KΩ nominal<br><b>A/3K Series:</b> 3KΩ nominal<br><b>A/AN Series (Type III):</b> 10KΩ nominal<br><b>A/AN-BC Series:</b> 5.238KΩ nominal<br><b>A/20K Series:</b> 20KΩ nominal<br><b>A/CP Series (Type II):</b> 10KΩ nominal<br><b>A/CSI Series:</b> 10KΩ nominal<br><b>A/10KS Series:</b> 10KΩ nominal<br><b>A/10K-E1 Series:</b> 10KΩ nominal<br><b>A/50K:</b> 50KΩ nominal   (Brown/Yellow)<br><b>A/100KS Series:</b> 100KΩ nominal |
| <b>Accuracy 0-70°C (32-158°F):</b>                      | +/- 0.2°C (+/- 0.36°F) except <b>A/10K-E1 Series:</b> +/- 0.3°C (+/- 0.54°F)<br><b>A/1.8K Series:</b> +/- 0.5°C @ 25°C (77°F) and +/- 1.0°C (+/- 1.8°F)                                                                                                                                                                                                                                                                                                      |
| <b>Stability:</b>                                       | Sensor Dependent; Contact ACI for more information on specific sensor                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Response Time (63% Step Change):</b>                 | 10 Seconds nominal                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Power Dissipation Constant:</b>                      | 3 mW/°C except <b>A/1.8K Series:</b> 1 mW/°C   <b>A/10K-E1 Series:</b> 2 mW/°C                                                                                                                                                                                                                                                                                                                                                                               |
| <b>LCD Display Supply Voltage:</b>                      | +9 to 35 VDC / 24 VAC (50/60 Hz)                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>LCD Display Supply Current/VA:</b>                   | < 4 mA / 0.12 VA                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>LCD Display Accuracy:</b>                            | +/- 2°F or +/- 2°C @ 71°F (21.5°C) Typical                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>LCD Display Descriptor   Number of Digits:</b>       | °F (Fahrenheit) or °C (Celsius)   3 1/2 Segment Display                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>LCD Display Life Expectancy:</b>                     | 50,000 Hours Minimum                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Set Point Specifications   Set Point Indication:</b> | See Ordering Grid options on back of Product Data Sheet                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Set Point Tolerance:</b>                             | +/- 10% of Range                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Override Option:</b>                                 | Short Thermistor (Default); Field (Jumper) Selectable “Dry Contact” Closure (Separate Input); Short Set Point available upon request                                                                                                                                                                                                                                                                                                                         |
| <b>Operating   Storage Temperature Range:</b>           | 1.5 to 50°C (35 to 122°F)   <b>Non-LCD:</b> -40 to 65°C (-40 to 149°F), <b>LCD Display:</b> -10 to 65°C (14 to 149°F)                                                                                                                                                                                                                                                                                                                                        |
| <b>Operating Humidity Range:</b>                        | 10 to 95% RH, non-condensing                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Connections   Wire Size:</b>                         | Screw Terminal Blocks   16 (1.31 mm²) to 26 (0.129 mm²) AWG                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Terminal Block Torque Rating:</b>                    | 0.5 Nm (Minimum); 0.6 Nm (Maximum)                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Enclosure Material   Color:</b>                      | “R2” Enclosure: ABS; Plastic, White, UL94-HB   “R” Enclosure: ABS Plastic, Beige UL94-HB                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Product Dimensions:</b>                              | See Drawing on back of Product Data Sheet                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Product Weight:</b>                                  | <b>A/XX-R/RS/RO Series:</b> 0.14 lbs. (63.5g)   <b>A/XX-RSO Series:</b> 0.18 lbs. (81.6g)<br><b>A/XX-R2/R2S/R2O Series:</b> 0.16 lbs. (72.6g)   <b>A/XX-R2SO Series:</b> 0.20 lbs. (90.7g)<br><b>All LCD Display Units:</b> 0.18 lbs. (81.6g)                                                                                                                                                                                                                |
| <b>Agency Approvals:</b>                                | CE*, RoHS2, WEEE                                                                                                                                                                                                                                                                                                                                                                                                                                             |

**Note\*\*:** All LCD Display Units are not CE Compliant, but they are RoHS2 Compliant

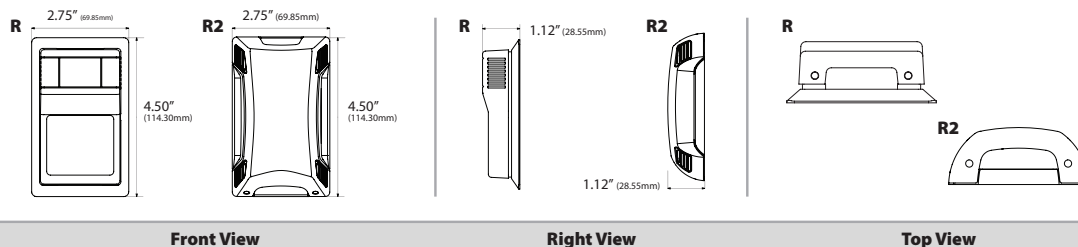


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TEMPERATURE | ##



## DIMENSIONAL DRAWING



## STANDARD ORDERING

Model # Example: **A/CP -OR- 144212**

| Model #         | Item # | Description                                     |
|-----------------|--------|-------------------------------------------------|
| <b>A/CP-R</b>   | 144212 | 10K (Type II), "R" Version, Beige, No Options   |
| <b>A/CP-R2</b>  | 144213 | 10K (Type II), "R2" Version, White, No Options  |
| <b>A/AN-R</b>   | 144208 | 10K (Type III), "R" Version, Beige, No Options  |
| <b>A/AN-R2</b>  | 144209 | 10K (Type III), "R2" Version, White, No Options |
| <b>A/20K-R</b>  | 144160 | 20K, "R" Version, Beige, No Options             |
| <b>A/20K-R2</b> | 144170 | 20K, "R2" Version, White, No Options            |

## CUSTOM ORDERING

Model # Example:

**A/** **1.8K** **RS** **RJ6** **LCD** **F** **G.** **A01** **B4**

## MODEL #

|                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |           |
|-------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>A. Sensor Series</b> No Selection Required                                                                           | <b>A/</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>A/</b> |
| <b>B. Model Series</b> Select One (1)                                                                                   | <b>1.8K</b>   <b>3K</b>   <b>AN</b>   <b>AN-BC</b>   <b>CP</b>   <b>CSI</b>   <b>10KS</b>   <b>10K-E1</b>   <b>20K</b>   <b>50K</b>   <b>100KS</b>                                                                                                                                                                                                                                                                                                                                                                                                                              |           |
| <b>C. Configuration</b> Select One (1)                                                                                  | <b>R</b> = Room   <b>RO</b> = Room with Override   <b>RS</b> = Room with Set Point<br><b>RSO</b> = Room with Set Point and Override   <b>R2</b> = Room   <b>R2O</b> = Room with Override<br><b>R2S</b> = Room with Set Point   <b>R2SO</b> = Room with Set Point and Override                                                                                                                                                                                                                                                                                                   |           |
| <b>D. Communication Jack</b> Select One (1)                                                                             | <b>----</b> = No Jack<br><b>RJ4</b> = 4 Pin 4 Conductor RJ9, RJ10, or RJ22 Style Head Set Modular Connector<br><b>RJ6</b> = 6 Pin 6 Conductor RJ12 Modular Phone Connector<br><b>232</b> = 3.5mm (1/8") Stereo Jack                                                                                                                                                                                                                                                                                                                                                             |           |
| <b>E. LCD Display</b> <sup>1</sup> Select One (1)                                                                       | <b>----</b> = No LCD Display   <b>LCD</b> <sup>1</sup> = With LCD Display (Only Available with "R" Style Enclosure)                                                                                                                                                                                                                                                                                                                                                                                                                                                             |           |
| <b>F. LCD Display Descriptor</b> Select One (1)                                                                         | <b>F</b> = °F (Fahrenheit)   <b>C</b> = °C (Celsius)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |           |
| <b>G. NIST</b> <sup>**</sup> Select One (1)                                                                             | <b>----</b> = No NIST Certificate   <b>NIST</b> = NIST Certificate (3 Points)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |           |
| <b>Setpoint Configuration Options</b> Select Options below if RS, RSO, R2S or R2SO was selected as a Configuration (C). |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |           |
| <b>1. Slidepots</b> <sup>***</sup> Select One (1)                                                                       | <b>Direct Acting</b> (Range in Ohms)<br><b>A01</b> = 0 to 100K   <b>A02</b> = 0 to 20K   <b>A03</b> = 0 to 10K   <b>A06</b> = 4.75K to 24.75K<br><b>A07</b> = 10K to 30K   <b>A08</b> = 1K to 11K   <b>A09</b> = 0 to 2K   <b>A10</b> = 0 to 1K<br><b>A11</b> = 2.05K to 3.05K   <b>A12</b> = 0 to 400   <b>A16</b> = 0 to 5K   <b>A18</b> = 10K to 15K<br><b>A20</b> = 6.19K to 26.19K   <b>A26</b> = 866 to 1,266   <b>A29</b> = 7.87K to 27.8K<br><br><b>Reverse Acting</b> (Range in Ohms)<br><b>A04</b> = 1051.1 to 51.1   <b>A14</b> = 10K to 0   <b>A24</b> = 9.5K to 1K |           |
| <b>2. Setpoint Stickers</b> Select One (1)                                                                              | <b>A3</b> = 18-28 DEG C   <b>A4</b> = 20-30 DEG C   <b>B4</b> = 55-85 DEG F   <b>B7</b> = 60-90 DEG F   <b>C5</b> = COOL/WARM<br><b>C6</b> = COOLER/WARMER   <b>D3</b> = WARM/COOL   <b>G5</b> = BLUE/RED (R2 Enclosure)                                                                                                                                                                                                                                                                                                                                                        |           |

**Note** <sup>1</sup>: LCD Display is not compatible with NIST | **Note** <sup>\*</sup>: Short Sensor is our default but the Dry Contact option is field selectable with terminals included  
**Note** <sup>\*\*</sup>: NIST is available in "R" and "R2" only configurations | **Note** <sup>\*\*\*</sup>: Other Setpoint configurations are available. Please contact ACI.

## ACCESSORIES ORDERING

Model # Example: **A/MOUNTING PLATE BEIGE R -OR- 106821**

| Model #                          | Item # | Description                                             |
|----------------------------------|--------|---------------------------------------------------------|
| <b>A/MOUNTING PLATE BEIGE R</b>  | 106821 | Wall Mounting Back Plate, Plastic, Beige ("R")          |
| <b>A/MOUNTING PLATE WHITE R</b>  | 126386 | Wall Mounting Back Plate, Plastic, White ("R")          |
| <b>A/MOUNTING PLATE WHITE R2</b> | 143369 | Wall Mounting Back Plate, Plastic, White ("R2")         |
| <b>LOCKING COVER</b>             | 107370 | Clear Thermostat Guard, Locking Cover, Low Profile      |
| <b>A/ROOM-FOAM-PAD</b>           | 125690 | 1/8" Foam Insulation Pad with Adhesive (3" x 2", Black) |





**Micromet**

Date:

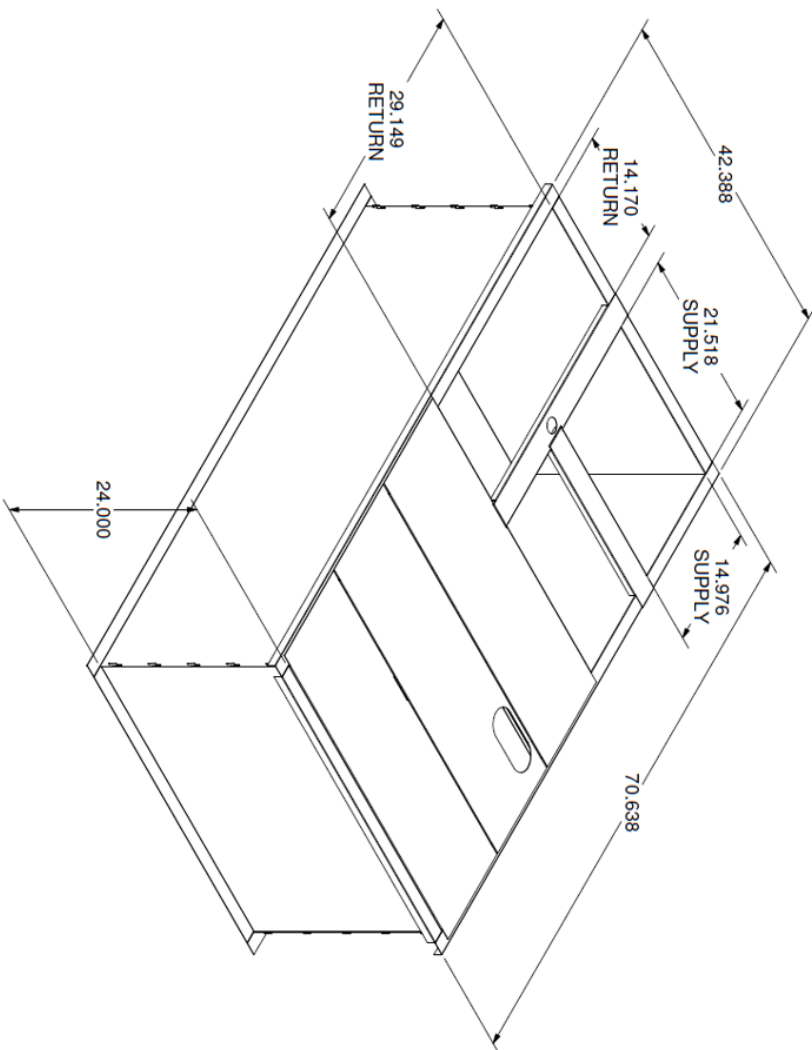
Specification Sheet for Daikin Small Cabinet 3 to 6 Ton  
24" Roof Curb

Part Number:  
Daikin PN: 0221L00015  
Micromet PN: CRBK-DKN12SA-2411

Submitted To:

Job:

Notes:



Notes:

1. Roof curb ships knocked down for field assembly. Package includes plastic corrugated end caps.
2. Curb includes wood nailer, attached to curb sides and ends, and gasket package shipped inside of curb package.
3. Deck pans include 1" – 1.5 lb. density insulation.
4. Curb sides and ends are constructed of 18 gauge galvanized steel.

**Product Operating Weight:**

109 Lbs

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**Micromet**

DATE:

WEIGHT (LBS):  
87.75

DESCRIPTION:  
SMALL VERTICAL ECONOMIZER

PART NUMBER:

ECD-DKN12CB-W00S  
0270L02827

SUBMITTED TO:

COMPANY:

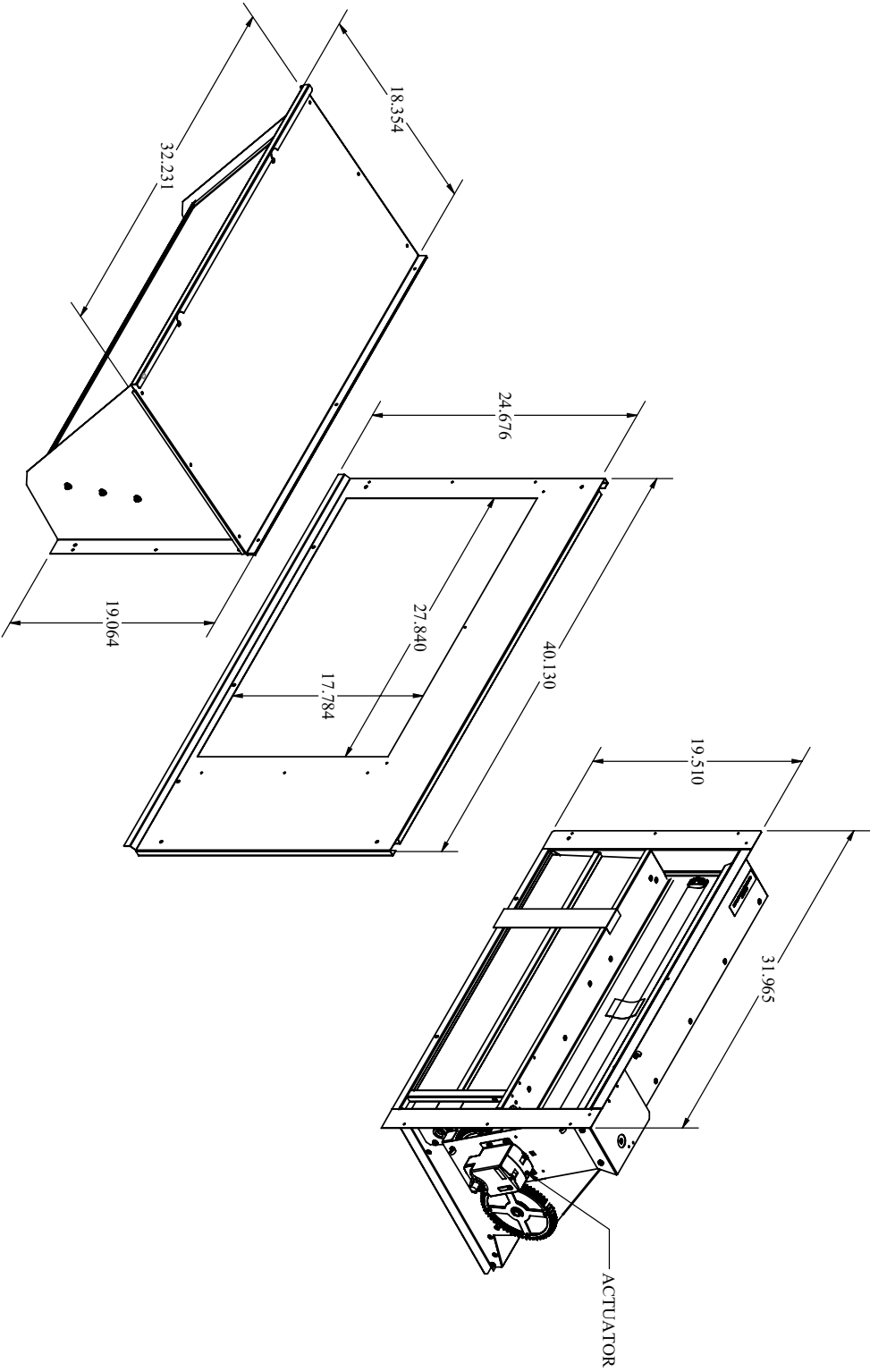
JOB NAME:

APPROVED BY:

DATE:

FOR MODEL NUMBER:

DETAILS:



NOTES:

CREATED BY:

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### ATTENTION INSTALLING PERSONNEL

As a professional installer you have an obligation to know the product better than the customer. This includes all safety precautions and related items.

Prior to actual installation, thoroughly familiarize yourself with this Instruction Manual. Pay special attention to all safety warnings. Often during installation or repair it is possible to place yourself in a position which is more hazardous than when the unit is in operation.

Remember, it is **your** responsibility to install the product safely and to know it well enough to be able to instruct a customer in its safe use.

Safety is a matter of common sense...a matter of thinking before acting. Most dealers have a list of specific good safety practices...follow them.

The precautions listed in this Installation Manual are intended as supplemental to existing practices. However, if there is a direct conflict between existing practices and the content of this manual, the precautions listed here take precedence.

### WARNING

#### HIGH VOLTAGE!

TO AVOID THE RISK OF ELECTRICAL SHOCK, WIRING TO THE UNIT MUST BE POLARIZED AND GROUNDED.



### WARNING

#### HIGH VOLTAGE!

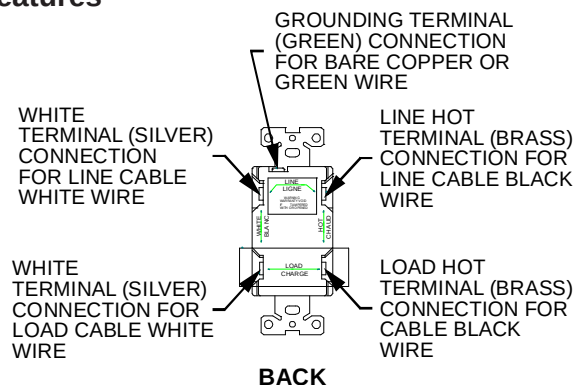
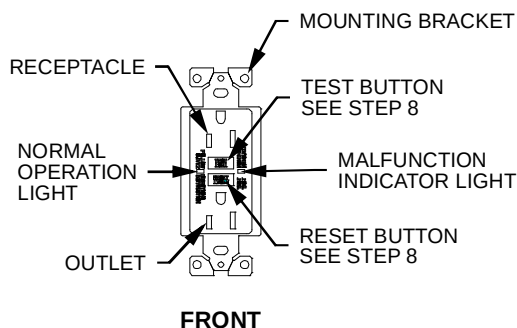
TO AVOID PERSONAL INJURY OR DEATH DUE TO ELECTRICAL SHOCK, DISCONNECT ELECTRICAL POWER BEFORE SERVICING OR CHANGING ANY ELECTRICAL WIRING.



### IMPORTANT!

- A GFCI receptacle is different from conventional receptacles. In the event of a ground fault, a GFCI will trip and quickly stop the flow of electricity to prevent serious injury.
- A GFCI receptacle does NOT protect against circuit overloads, short circuits, or shocks. For example, you can still be shocked if you touch bare wires while standing on a non-conducting surface such as a wood floor.
- Your unit has been built with either a powered or non-powered 115V / 15A GFCI outlet for your convenience. This is to only be used while servicing the unit.
- Use this GFCI with copper or copper-clad wire. Do not use it with aluminum wire.
- Do not install this GFCI receptacle on a circuit that powers life support equipment because if the GFCI trips it will shut down the equipment.
- The GFCI receptacle must be kept dry by a weather proof cover and box at all times.
- Test the outlet before use to make sure it has not incurred any damage during shipping. Perform the *Operation Check* listed in this manual.

### GFCI Features



SCREW (TERMINAL) COLORS:  
GREEN = GROUNDING TERMINAL  
SILVER = WHITE TERMINALS  
BRASS = HOT TERMINALS



Your unit will have either a **powered** or **non-powered** GFCI outlet. If your unit has a factory-installed powered convenience outlet, the outlet is ready for use once the unit is installed and operating. See figure below for component locations of the powered outlet.



If the power for either needs to be altered from 230V to 208V, please review and follow the instructions listed on the outlet wiring diagram next to the unit diagram. Ensure unit power is OFF before making any wiring changes. See diagram below for your convenience.



**NOTE:** There is a 15A breaker mounted to the indoor section of the unit. If outlet is not powered, check the breaker to make sure it is in the ON position. If the breaker has been tripped, check for shorts in the wiring.

## NON-POWERED GFCI OUTLET INSTALLATION INSTRUCTIONS

1. Turn power OFF to unit and power source used to power the outlet.
2. Locate the outlet on your unit. It will be near the control box area.
3. Remove gasket and outlet.
4. Bring power into the outlet box as shown. Ensure the wiring is suitable for outdoor conditions. Connect the white wire to the WHITE terminal (Silver).
5. Connect the black wire to the HOT terminal (Brass).
6. Connect a 6-inch bare copper (or GREEN) 12 or 14 AWG wire to the grounding terminal on the GFCI, and a similar wire to the grounding terminal on the box. Connect the ends of these wires to the LINE cable's bare copper (or GREEN) wire using a wire connector.
7. Fold the wires into the box, keeping the grounding wire away from the WHITE and HOT terminals. Screw the receptacle to the box.
8. Install cover as per instructions.



## OPERATION CHECK

1. Turn power ON. Press the RESET button fully. Plug a lamp or radio into the GFCI (and leave it plugged-in) to verify that the power is ON.
2. Next, press the TEST button to trip the device. The radio or lamp should shut OFF and the RESET button will pop-out. To restore power, press the RESET button.
3. If the malfunction indicator light (end-of-life indicator LED) (red) lights, the GFCI has stopped functioning and needs to be replaced.

If the outlet does not work as described, turn power OFF and check the wire connections against the diagram shown above. Ensure that there are not loose wires or loose connections.

If the RESET button does not work, check the LINE and LOAD connections. They could be reversed. If there are no wiring mistakes, the receptacle could be damaged and will need to be replaced. If the connections are rewired to the GFCI, please perform the *Operation Check* again.

**NOTE:** There is a 15A breaker mounted to the indoor section of the unit. If outlet is not powered, check the breaker to make sure it is in the ON position. If the breaker has been tripped, check for shorts in the wiring.

Check the TEST and RESET buttons monthly to assure proper operation.

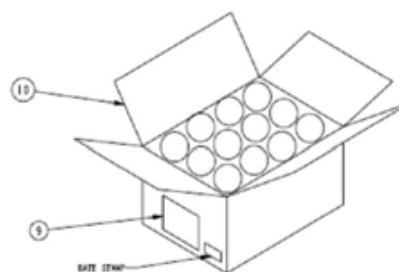
## NON-POWERED AND POWERED GFCI OUTLET COVER INSTALLATION INSTRUCTIONS

1. Locate the outlet on your unit. It will be near the control box area.
2. Remove the gasket.
3. Install the GFCI outlet cover as shown.
4. The outlet cover is shipped with the unit and could be found in control box or blower deck.

**ASSEMBLY INSTRUCTIONS:** Numbers in parenthesis reflect the balloon numbers on the drawing.

1. Apply gasket (ITEM 6) to base (ITEM 2).
2. Apply adhesive, LOCTITE 4204 or 4205, (ITEM 13) into cord flap gasket slot of cover (ITEM 1) and base (ITEM 2). See gasket slot detail.

3. Insert cord flaps (ITEM 12) into slots of cover and base within 15 seconds of step 2 above. Ensure cord flaps are fully seated and top of flap is flush with the top of cord opening of cover and base. See cord opening detail. Presence of adhesive should not be apparent on any exposed surface of installed cord flap gasket.
4. Snap cover (ITEM 1) to base (ITEM 2).
5. Place wallplates (ITEM 3 & 4) into assembled unit.
6. Place bagged mounting screws (ITEM 5) into assembled unit.
7. Place instruction sheet (ITEM 8) into assembled unit.
8. Close assembled unit.
9. Place assembled unit into packaging bag (ITEM 7).
10. Seal packaging bag (ITEM 7).
11. Place sealed bags into carton (ITEM 10), four rows of three.
12. Place label (ITEM 9) on carton (ITEM 10).
13. Insert quality flyer (ITEM 11).



NOTE: SPECIFICATIONS AND PERFORMANCE DATA LISTED HEREIN ARE SUBJECT TO CHANGE WITHOUT NOTICE

### Quality Makes the Difference!

All of our systems are designed and manufactured with the same high quality standards regardless of size or efficiency. We have designed these units to significantly reduce the most frequent causes of product failure. They are simple to service and forgiving to operate. We use quality materials and components. Finally, every unit is run tested before it leaves the factory. That's why we know. . . **There's No Better Quality.**

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|                     |                     |
|---------------------|---------------------|
| Project Name: _____ | Approval: _____     |
| Location: _____     | Date: _____         |
| Engineer: _____     | Construction: _____ |
| Submitted to: _____ | Unit #: _____       |
| Submitted by: _____ | Drawing #: _____    |
| Reference: _____    |                     |

## MODEL COMPATIBILITY:

Compatible with Light Commercial models: 3 to 12.5 ton High Efficiency (DR\* series), 15 to 25 ton Standard Efficiency (DB\* series)

## SPECIFICATIONS:

|                                   |                                                                                                                            |                                                  |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| <b>Model</b>                      | 0130M00579                                                                                                                 | 0130M00580                                       |
| <b>Description</b>                | Daikin iLINQ DDC Controller - Standard                                                                                     | Daikin iLINQ DDC Controller - with MHGRH Control |
| <b>Dimensions</b>                 | 12.4 x 5.2 x 2.8 inches                                                                                                    |                                                  |
| <b>Operating Conditions</b>       | -4°F to 140°F, 90% RH non-condensing                                                                                       |                                                  |
| <b>Storage Conditions</b>         | -22°F to 158°F, 90% RH non-condensing                                                                                      |                                                  |
| <b>Supply Voltage</b>             | Dedicated class 2 transformer<br>24VAC (+10/-15%), 45 VA                                                                   |                                                  |
| <b>Wiring specs</b>               | 18 to 22 AWG twisted pair or shielded cable for all sensor installations                                                   |                                                  |
| <b>Temperature Inputs</b>         | Type III thermistor, 10K $\Omega$ @ 77°F                                                                                   |                                                  |
| <b>Analog Inputs (AI)</b>         | 0 – 10VDC, Input Precision $\pm 0.3\%$ Full Scale                                                                          |                                                  |
| <b>Number of AIs</b>              | 8                                                                                                                          | 10                                               |
| <b>Digital Inputs (DI)</b>        | 24VAC (+10/-15%) 50/60HZ, Absorbed Current: 5mA                                                                            |                                                  |
| <b>Number of DIs</b>              | 14                                                                                                                         | 18                                               |
| <b>Analog Outputs (AO)</b>        | 0 – 10Vdc, External Power Supply: 24VAC (+10/-15%), Precision: $\pm 2\%$ Full Scale, Resolution: 8 Bit, Maximum Load: 10mA |                                                  |
| <b>Number of AOs</b>              | 4                                                                                                                          | 6                                                |
| <b>Digital Outputs (DO)</b>       | 8A/250VAC Resistive Load                                                                                                   |                                                  |
| <b>Number of DOs</b>              | 13                                                                                                                         | 18                                               |
| <b>USB Type “B” Port</b>          | 1 for Programming, Setup, and Advanced Diagnostics                                                                         |                                                  |
| <b>USB Type “A” Port</b>          | 1 for Transferring files                                                                                                   |                                                  |
| <b>Ethernet Ports</b>             | 2 autocross 10/100 MBPS port for BACnet/IP communication and Web Interface                                                 |                                                  |
| <b>BACnet/IP Communication</b>    | Built-in BACnet/IP (Ethernet port)                                                                                         |                                                  |
| <b>BACnet MS/TP Communication</b> | Built-in RS485 port, Baud rate select from 9600, 19200, 38400 (Default), 57600, and 76800                                  |                                                  |
| <b>LonWorks Communication</b>     | Optional communication card (0130M00584) required                                                                          |                                                  |
| <b>Compliance</b>                 | UL Listed / CUL Listed<br>BTL Certified (B-BC)<br>California Title 24 certified economizer control                         |                                                  |

## PRODUCT IMAGE:



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# Submittal Data Sheet

0130M00579/0130M00580– Daikin iLINQ DDC  
Controller

|               |  |               |  |
|---------------|--|---------------|--|
| Project Name: |  | Approval:     |  |
| Location:     |  | Date:         |  |
| Engineer:     |  | Construction: |  |
| Submitted to: |  | Unit #:       |  |
| Submitted by: |  | Drawing #:    |  |
| Reference:    |  |               |  |

## OPTIONS:

| Type    | Name                                        | Part Number | Function                                                                                                                                                                                                                                                                                                                                                                          |
|---------|---------------------------------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Module  | MHGRH Expansion Module                      | 0130M00581  | Used for 0130M00580 only<br>DDC Controller expansion module for control of the Modulating Hot Gas Reheat (MHGRH) valve                                                                                                                                                                                                                                                            |
|         | LonWorks Communication Card                 | 0130M00584  | Communication card allowing connection of the DDC Controller to a LonWorks network                                                                                                                                                                                                                                                                                                |
| Sensors | Space CO <sub>2</sub> Sensor                | 0130L00225  | Sensor installed on wall to monitor space CO <sub>2</sub> level. A CO <sub>2</sub> sensor is required for demand control ventilation. Sensor output range 0-10Vdc, 0-2000ppm                                                                                                                                                                                                      |
|         | Duct CO <sub>2</sub> Sensor                 | 0130L00220  | Sensor installed in return air duct to monitor space CO <sub>2</sub> level. A CO <sub>2</sub> sensor is required for demand control ventilation. Sensor output range 0-10Vdc, 0-2000ppm                                                                                                                                                                                           |
|         | Duct Temperature Sensor 4"                  | 0130L00222  | Sensor installed in return air or supply air duct to monitor temperature. When installed in the return air duct, this sensor can be used in place of the wall mounted temperature sensor. Sensor type is 10KΩ Type III Thermistor                                                                                                                                                 |
|         | Duct Temperature Sensor 8"                  | 0130L00228  |                                                                                                                                                                                                                                                                                                                                                                                   |
|         | Duct Temperature and Humidity Sensor        | 0130L00221  | Sensor installed in return air duct to monitor space temperature and humidity level. This sensor can be used in place of the wall mounted temperature and humidity sensor. Temperature sensor type 10KΩ Type III Thermistor. Humidity sensor range 0-10Vdc, 0-100%RH                                                                                                              |
|         | Outdoor Air Temperature Sensor              | 0130L00227  | Sensor installed to monitor outdoor air temperature. Sensor type 10KΩ Type III Thermistor                                                                                                                                                                                                                                                                                         |
|         | Outdoor Air Temperature and Humidity Sensor | 0130L00223  | Sensor installed to monitor outdoor temperature and humidity levels. This sensor can be used in place of the outdoor temperature sensor when humidity must also be monitored. Temperature sensor type 10KΩ Type III Thermistor. Humidity sensor range 0-10Vdc, 0-100%RH                                                                                                           |
|         | Space Temperature Sensor                    | 0130L00226  | Sensor installed on wall to monitor space temperature. This sensor also provides local setpoint adjustment and a temporary occupancy override button. Sensor type 10KΩ Type III Thermistor                                                                                                                                                                                        |
|         | Space Temperature and Humidity Sensor       | 0130L00224  | Sensor installed on wall to monitor space temperature and humidity levels. This sensor can be used in place of the wall mounted temperature sensor when humidity must also be monitored. This sensor also provides local setpoint adjustment and a temporary occupancy override button. Temperature sensor type 10KΩ Type III Thermistor. Humidity sensor range 0-10Vdc, 0-100%RH |

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0130M00579/0130M00580– Daikin iLINQ DDC  
Controller

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| Location:     |  | Date:         |  |
| Engineer:     |  | Construction: |  |
| Submitted to: |  | Unit #:       |  |
| Submitted by: |  | Drawing #:    |  |
| Reference:    |  |               |  |

## FEATURES:

- Packaged RTU control:
  - Single speed, two speed, and variable (5) speed blower configurations
  - Blower proving switch software interlock
  - 1 or 2 stages of heating with PID control load calculation
  - Gas, electric stages, electric with SCR control, and heat pump heating configurations
  - 1 or 2 stages of cooling with PID control load calculation
  - 1, 2 or 4 compressors with pressure switch feedback and alarms
  - Lead / lag compressor priority rotation based on runtime
  - Independent defrost of condenser coils
  - Demand defrost interval calculation
- Staged auxiliary electric heat during defrost or when heat pump heating is locked out
- Dehumidification using Modulating Hot Gas Reheat (MHGRH)
- Low suction pressure freeze protection on units with MHGRH dehumidification
- Low ambient condenser fan control on units with MHGRH dehumidification
- Demand control ventilation
- Exhaust fan enable
- Dirty filter alarm
- Emergency shutdown interlock and alarm
- Remote start/stop
- Load shedding
- Local time scheduling, including weekly and holiday events
- Automatic daylight savings time adjustment
- Optimal start / optimal stop
- Onboard trend log storage can be exported to .csv file for analysis
- Live trend logs viewable via web interface
- Selectable TSTAT mode allowing for connection to standard TSTAT, bypassing some of the DDC control logic
- BACnet MS/TP or BACnet IP communication
- LonWorks communication with optional field installed module
- Web interface for commissioning or monitoring through any web browser
- Onboard LCD display for local commissioning or monitoring

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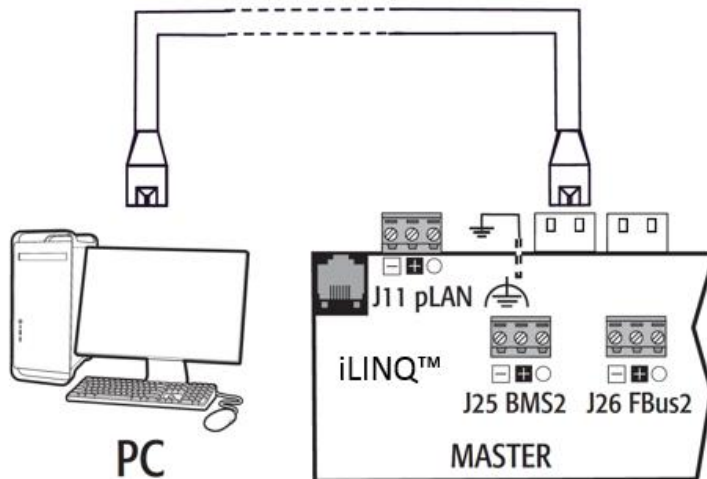
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|                     |                     |
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| Submitted to: _____ | Unit #: _____       |
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| Reference: _____    |                     |

## SYSTEM DIAGRAM:

- Ethernet Network (Web Interface or BACnet/IP Communication):
  - The controller can be connected to a network switch or PC individually



- Multiple controller can be daisy-chained together to a network switch or PC utilizing the second Ethernet port.



- BACnet MS/TP Network

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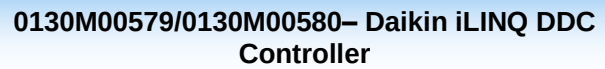
- LonWorks Network



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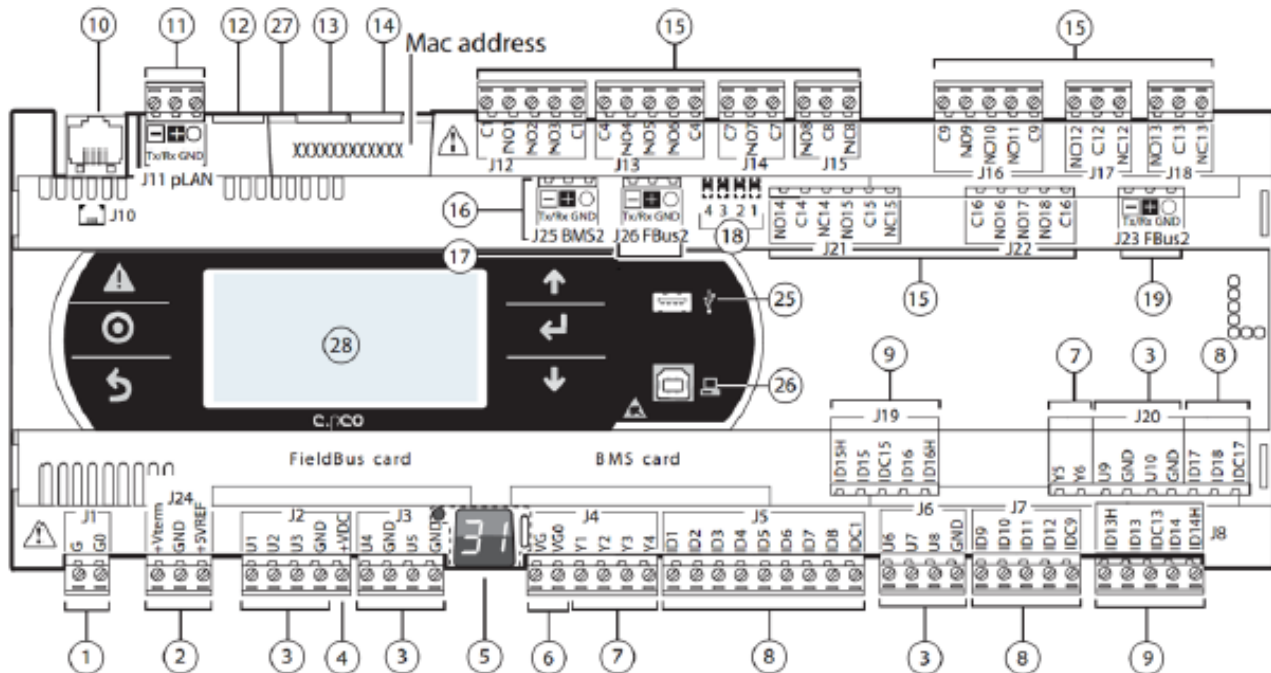
Drawing #:

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Project Name: \_\_\_\_\_  
 Location: \_\_\_\_\_  
 Engineer: \_\_\_\_\_  
 Submitted to: \_\_\_\_\_  
 Submitted by: \_\_\_\_\_  
 Reference: \_\_\_\_\_

Approval: \_\_\_\_\_  
 Date: \_\_\_\_\_  
 Construction: \_\_\_\_\_  
 Unit #: \_\_\_\_\_  
 Drawing #: \_\_\_\_\_

• IO Layout Large (0130M00580):



| REF | DESCRIPTION                                                      | REF | DESCRIPTION             |
|-----|------------------------------------------------------------------|-----|-------------------------|
| 1   | Power Connector [G(+), G0(-)]                                    | 13  | Ethernet Port 1         |
| 2   | +VTERM: Terminal Power Supply<br>+5VREF: 5VDC Probe Power Supply | 14  | Ethernet Port 2         |
| 3   | Analog Inputs                                                    | 15  | Relay Outputs           |
| 4   | +VDC: 24VDC Power For Active Probes                              | 16  | BMS Port                |
| 5   | pLAN Address LED                                                 | 17  | Fieldbus Port           |
| 6   | 24VAC Power Input For Analog Outputs                             | 18  | Fieldbus/BMS Jumpers    |
| 7   | Analog Outputs                                                   | 19  | Fieldbus Port           |
| 8   | Digital Inputs                                                   | 25  | USB Host Port           |
| 9   | Digital Inputs                                                   | 26  | USB Device Port         |
| 10  | pLAN Connection For Room Terminal                                | 27  | Earth Ground Connection |
| 11  | pLAN Connection For Room Terminal                                | 28  | Display and Keypad      |
| 12  | Reserved                                                         |     |                         |

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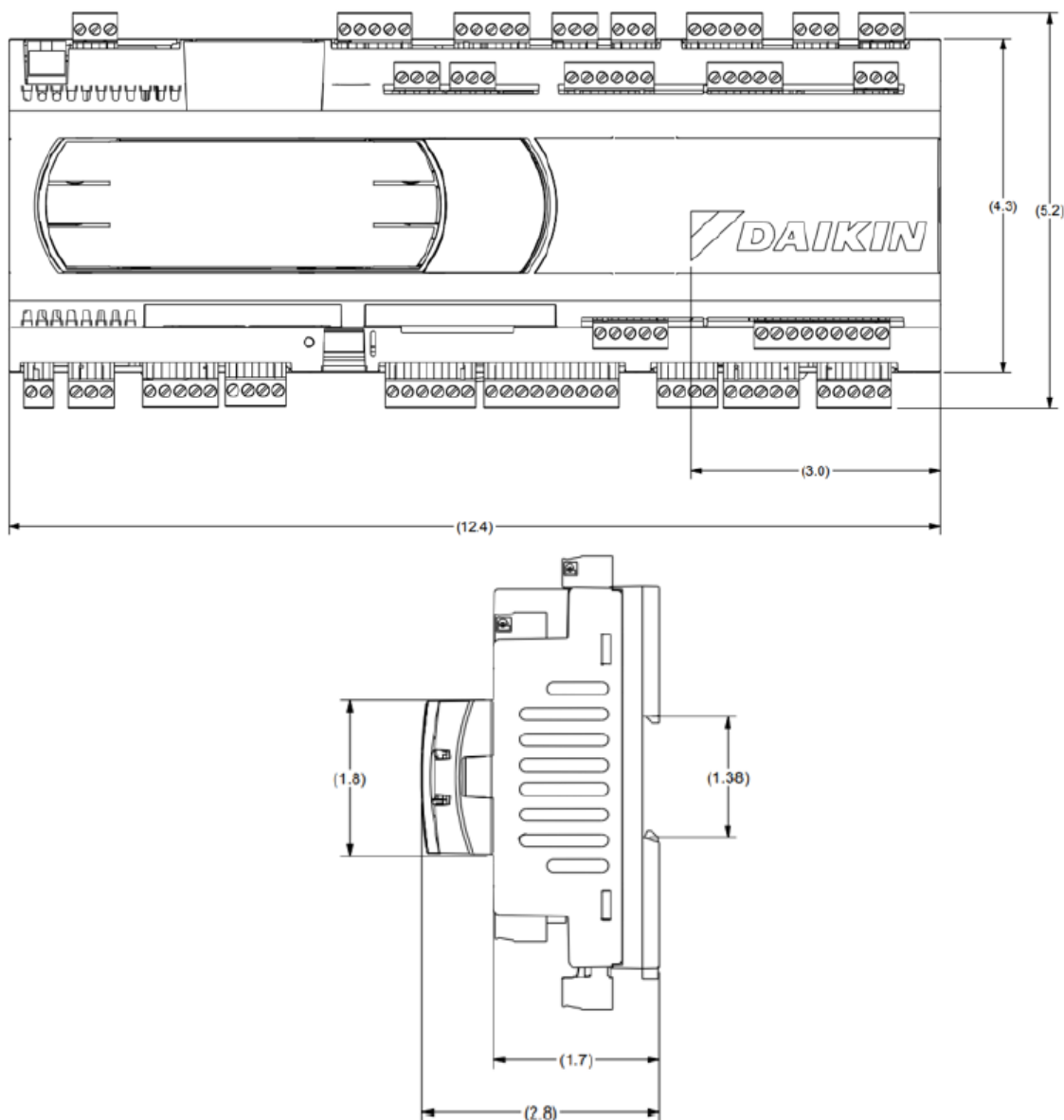
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| Submitted by: _____ | Drawing #: _____    |
| Reference: _____    |                     |

## DIMENSIONS:

The physical dimension for the medium and large controllers are the same (inches):



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Controller

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| Submitted to: |  | Unit #:       |  |
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| Reference:    |  |               |  |

## DOCUMENTATION:

Documentation available on [www.daikincity.com](http://www.daikincity.com) and/or [www.daikinac.com](http://www.daikinac.com):

- User Manual
- Quick Start Guide
- BACnet Design Guide
- LonWorks Design Guide
- Submittal
- Product Flyer

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# FACTORY INSTALLED ELECTRICAL DISCONNECTS INSTALLATION INSTRUCTIONS

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www.daikinac.com  
P/N: IOD-7002A Date: JULY 2019

## DESCRIPTION

This Installation and Operation manual covers Factory Installed Electrical Disconnects for Daikin Commercial Package Units.



### WARNING

**ONLY PERSONNEL THAT HAVE BEEN TRAINED TO INSTALL, ADJUST, SERVICE OR REPAIR (HEREINAFTER, "SERVICE") THE EQUIPMENT SPECIFIED IN THIS MANUAL SHOULD SERVICE THE EQUIPMENT. THE MANUFACTURER WILL NOT BE RESPONSIBLE FOR ANY INJURY OR PROPERTY DAMAGE ARISING FROM IMPROPER SERVICE OR SERVICE PROCEDURES. IF YOU SERVICE THIS UNIT, YOU ASSUME RESPONSIBILITY FOR ANY INJURY OR PROPERTY DAMAGE WHICH MAY RESULT. IN ADDITION, IN JURISDICTIONS THAT REQUIRE ONE OR MORE LICENSES TO SERVICE THE EQUIPMENT SPECIFIED IN THIS MANUAL, ONLY LICENSED PERSONNEL SHOULD SERVICE THE EQUIPMENT. IMPROPER INSTALLATION, ADJUSTMENT, SERVICING OR REPAIR OF THE EQUIPMENT SPECIFIED IN THIS MANUAL, OR ATTEMPTING TO INSTALL, ADJUST, SERVICE OR REPAIR THE EQUIPMENT SPECIFIED IN THIS MANUAL WITHOUT PROPER TRAINING MAY RESULT IN PRODUCT DAMAGE, PROPERTY DAMAGE, PERSONAL INJURY OR DEATH.**

## ATTENTION INSTALLING PERSONNEL

As a professional installer, you have an obligation to know the product better than the customer. This includes all safety precautions and related items.

Prior to actual installation, thoroughly familiarize yourself with this Instruction Manual. Pay special attention to all safety warnings. Often during installation or repair, it is possible to place yourself in a position which is more hazardous than when the unit is in operation.

Remember, it is your responsibility to install the product safely and to know it well enough to be able to instruct a customer in its safe use.

Safety is a matter of common sense...a matter of thinking before acting. Most dealers have a list of specific good safety practices...follow them.

The precautions listed in this Installation Manual are intended as supplemental to existing practices. However, if there is a direct conflict between existing practices and the content of this manual, the precautions listed here take precedence.



### WARNING

#### HIGH VOLTAGE!

**TO AVOID THE RISK OF ELECTRICAL SHOCK, WIRING TO THE UNIT MUST BE POLARIZED AND GROUNDED.**



## PANEL / HANDLE INSTALLATION INSTRUCTIONS

**NOTE:** All field wiring must be per the National Electric Code. Use Copper Conductors only.



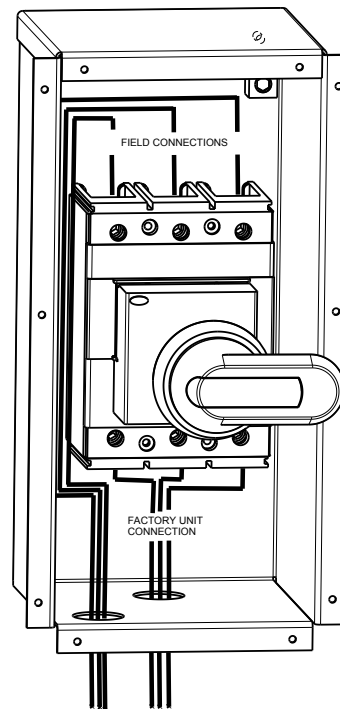
### WARNING

#### HIGH VOLTAGE!

**DISCONNECT ALL POWER BEFORE SERVICING OR INSTALLING THIS UNIT. MULTIPLE POWER SOURCES MAY BE PRESENT. FAILURE TO DO SO MAY CAUSE PROPERTY DAMAGE, PERSONAL INJURY OR DEATH.**



1. Remove disconnect panel from unit making sure to keep the panel mounting screws.
2. Remove "label" that is covering the Disconnect Shaft opening.
3. Route field wiring through the wire inlet and connect to the top lugs of the disconnect switch (Figure 1). **NOTE:** Disconnect switches are functional in any position. Some might be mounted "upside down". Field wiring should connect in the top as shown.



**Figure 1**

**NOTE:** For illustration purposes ONLY.  
Unit may not be exactly as shown.



4. Attach Handle Assembly to Disconnect Panel with handle oriented as shown in Figure 2. Make sure the handle is in the "OFF" position.



**Figure 2**

6. Align shaft with the opening in the Handle Assembly and slide Panel/Handle Assembly into place.
7. Secure the panel with the panel mounting screws (Figure 4).



**Figure 4**

5. Insert "shaft" into the opening on the DISCONNECT SWITCH (Figure 3). Make sure switch is in the "OFF" position (rotated fully counter-clockwise). The "pin" at the handle end of the shaft will be in the horizontal position.

**NOTE:** Some shafts have a "chamfer" along one edge. This chamfer should be oriented to the "bottom left" when the switch is in the "OFF" position.



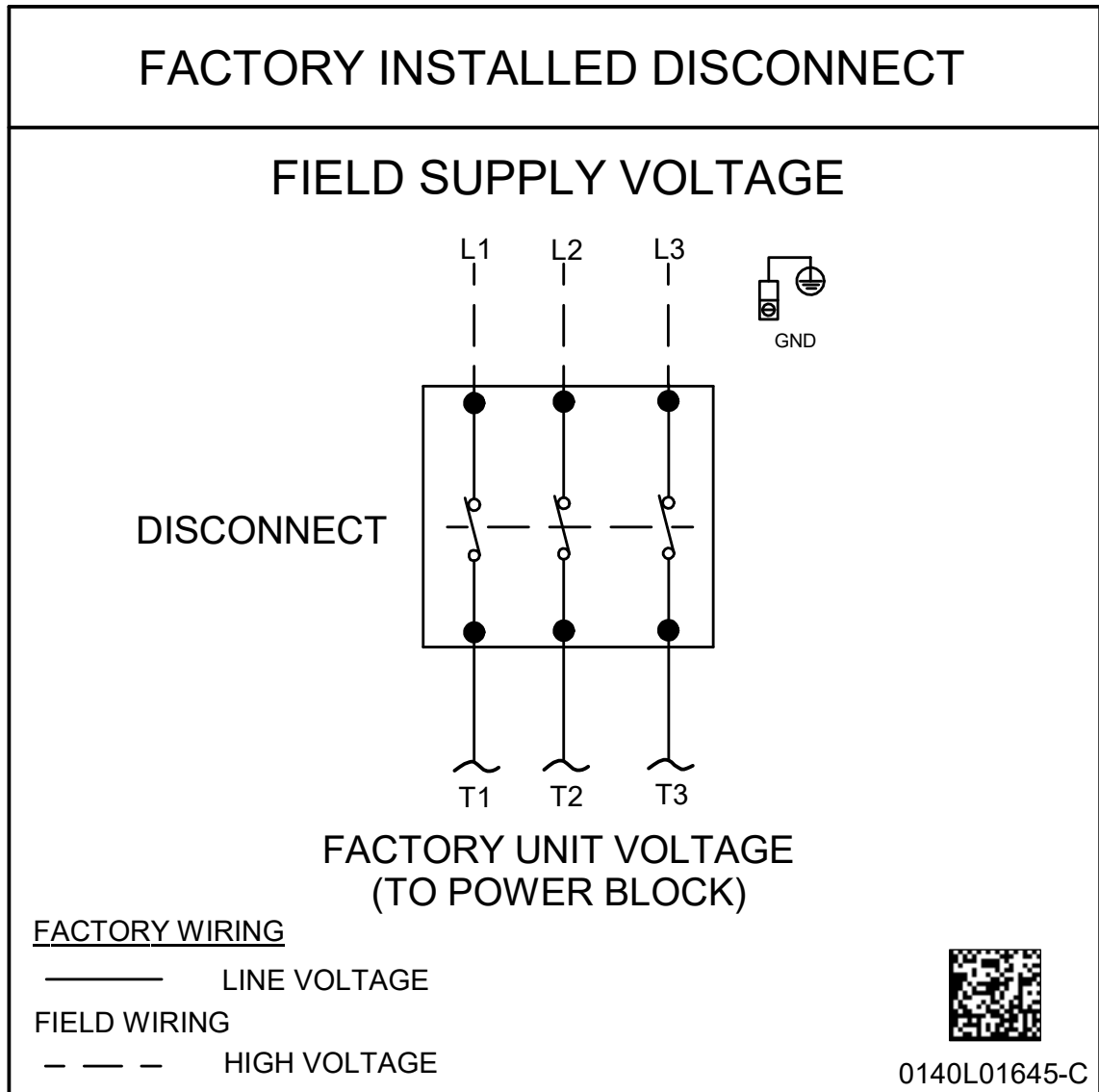
**Figure 3**

## APPENDIX A WIRING DIAGRAM

**HIGH VOLTAGE!**  
DISCONNECT ALL POWER BEFORE SERVICING OR INSTALLING THIS UNIT. MULTIPLE POWER SOURCES MAY BE PRESENT. FAILURE TO DO SO MAY CAUSE PROPERTY DAMAGE, PERSONAL INJURY OR DEATH.

**WARNING**





Wiring is subject to change. Always refer to the wiring diagram on the unit for the most up-to-date wiring.

#### CUSTOMER FEEDBACK

Daikin is very interested in all product comments.

Please fill out the feedback form on the following link:

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You can also scan the QR code on the right to be directed to the feedback page.



**NOTE: SPECIFICATIONS AND PERFORMANCE DATA LISTED HEREIN ARE SUBJECT TO CHANGE WITHOUT NOTICE**

#### **Quality Makes the Difference!**

All of our systems are designed and manufactured with the same high quality standards regardless of size or efficiency. We have designed these units to significantly reduce the most frequent causes of product failure. They are simple to service and forgiving to operate. We use quality materials and components. Finally, every unit is run tested before it leaves the factory. That's why we know. . . **There's No Better Quality.**

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## HAIL GUARDS

### FOR 3 - 12.5 TON LIGHT

### COMMERCIAL ROOFTOP UNITS

### MODELS: DRC/DRG/DRH/DBC/DBG/DBH

### INSTALLATION INSTRUCTIONS

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[www.daikincomfort.com](http://www.daikincomfort.com)

P/N: IOD-7083B Date: November 2019



#### WARNING

Only personnel that have been trained to install, adjust, service or repair (hereinafter, "service") the equipment specified in this manual should service the equipment. The manufacturer will not be responsible for any injury or property damage arising from improper service or service procedures. If you service this unit, you assume responsibility for any injury or property damage which may result. In addition, in jurisdictions that require one or more licenses to service the equipment specified in this manual, only licensed personnel should service the equipment.

Improper installation, adjustment, servicing or repair of the equipment specified in this manual, or attempting to install, adjust, service or repair the equipment specified in this manual without proper training may result in product damage, property damage, personal injury or death.

To select proper hail guard please use Hail Guard selection chart.

Hail Guard selection chart

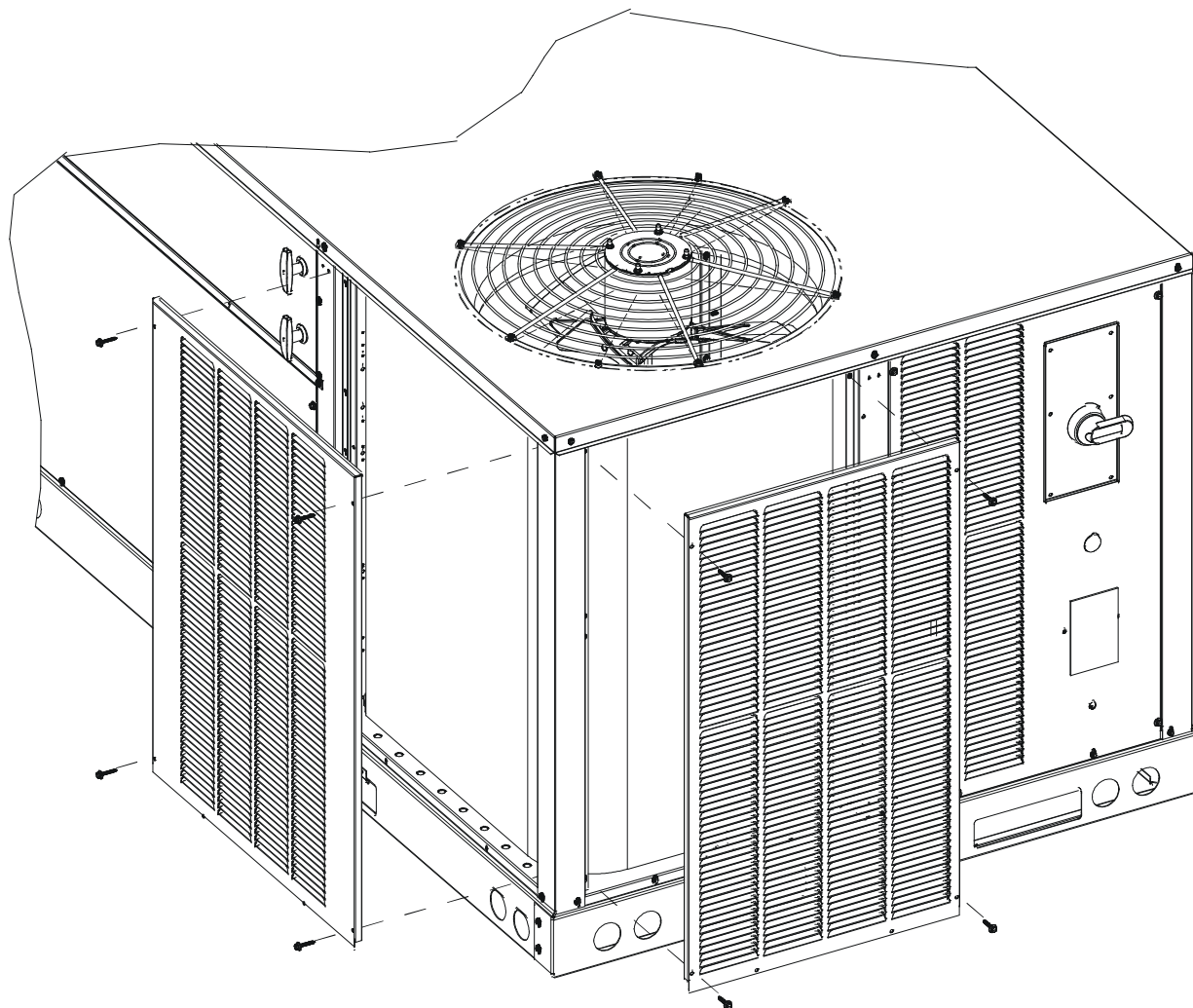
| Model  | Kit Part Number |                |             |                |             |
|--------|-----------------|----------------|-------------|----------------|-------------|
|        | HAILGD036HE     | HAILGD048060HE | HAILGD072HE | HAILGD090120HE | HAILGD150HE |
| DRC036 | X               |                |             |                |             |
| DRG036 | X               |                |             |                |             |
| DRH036 |                 | X              |             |                |             |
| DR*048 |                 | X              |             |                |             |
| DR*060 |                 | X              |             |                |             |
| DR*072 |                 |                | X           |                |             |
| DR*090 |                 |                |             | X              |             |
| DR*102 |                 |                |             | X              |             |
| DR*120 |                 |                |             | X              |             |
| DR*150 |                 |                |             |                | X           |
| DB*036 | X               |                |             |                |             |
| DB*048 | X               |                |             |                |             |
| DBC060 | X               |                |             |                |             |
| DBG060 | X               |                |             |                |             |
| DBH060 |                 | X              |             |                |             |
| DBG072 |                 | X              |             |                |             |
| DB*090 |                 |                |             | X              |             |
| DB*102 |                 |                |             | X              |             |
| DB*120 |                 |                |             | X              |             |
| DBC150 |                 |                |             | X              |             |
| DBG150 |                 |                |             | X              |             |
| DBH150 |                 |                |             |                | X           |

\* stands for C, G or H

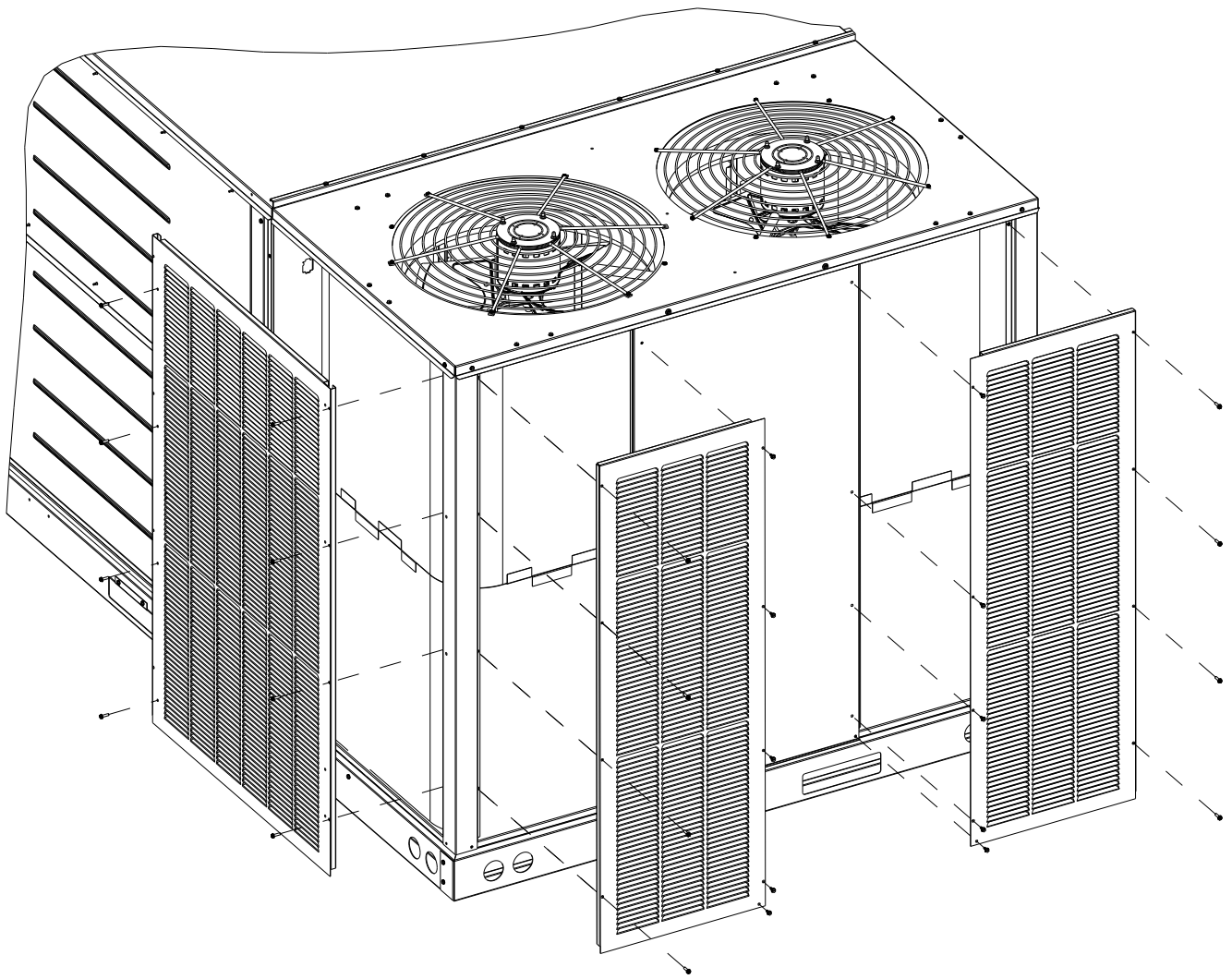


**Hail Guards are to be installed on the outside of the condenser coils, see figures 1 and 2.**

For installation, slide the top of the Hail Guard up and under the Top Panel flange, then push the bottom of the Hail Guard into position. With one hand holding guard up to the unit, put one screw into each corner of the guard to hold it onto the unit. Do the same for remaining guards. For installation of Hail Guard see Figure 1 and Figure 2. Use screws supplied with kit. Do not use longer screws - longer screws may damage the coil.



**Hail Guards Installation Figure 1**



**Hail Guards Installation Figure 2**

#### **CUSTOMER FEEDBACK**

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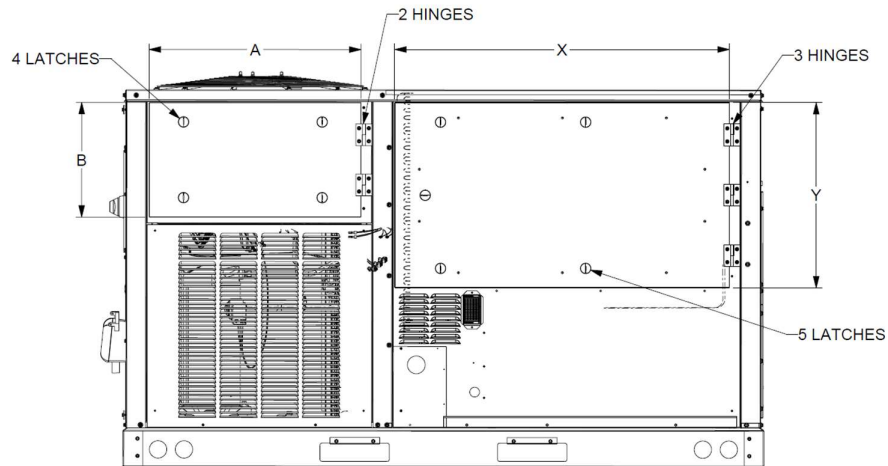
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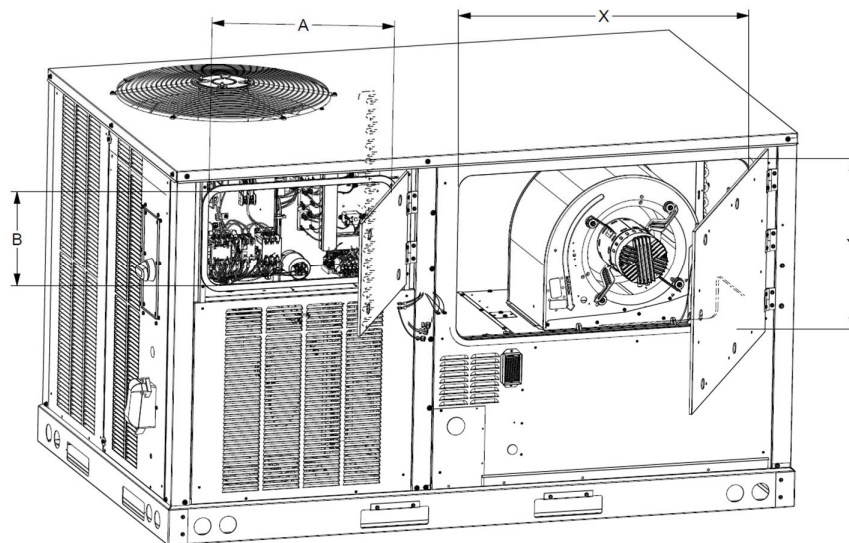
## HINGED DOOR, 3-6 TON

Factory installed hinged door option for 3-6 ton Light Commercial models.



➤ Hinged door size: (All dimensions in inches)

- 3/4/5/6 Ton: A=24.46, B=13.22, X=38.63
- Y=17.82 (DS\* 3/4/5 Ton & DHC/DHG 3 Ton), 21.39 (DS\* 6 Ton; DHH 3 Ton, DH\* 4/5 Ton), 31.67 (DH\* 6 Ton)



➤ Blower and control box opening: (All dimensions in inches)

- 3/4/5/6 Ton: A=21.67, B=10.92, X=34.25
- Y=15.54 (DS\* 3/4/5 Ton & DHC/DHG 3 Ton), 19.10 (DS\* 6 Ton; DHH 3 Ton, DH\* 4/5 Ton), 29.39 (DH\* 6 Ton)

# TTBCKHE01 & TTBCKHE02 THRU-THE-BOTTOM SERVICE CONNECTION KIT (FOR 3-12.5 Ton) LIGHT COMMERCIAL GAS UNITS INSTALLATION INSTRUCTIONS

© 2019 **DAIKIN MANUFACTURING COMPANY, L.P.**

19001 Kermier Rd. Waller, TX 77484

www.daikincomfort.com

P/N: IOD-7118 Date: October 2019

## Safety Considerations

The following symbols and labels are used throughout this manual to indicate immediate or potential safety hazards. It is the owner's and installer's responsibility to read and comply with all safety information and instructions accompanying these symbols. Failure to heed safety information increases the risk of personal injury, property damage, and/or product damage.

 **WARNING**

**ONLY PERSONNEL THAT HAVE BEEN TRAINED TO INSTALL, ADJUST, SERVICE OR REPAIR (HEREINAFTER, "SERVICE") THE EQUIPMENT SPECIFIED IN THIS MANUAL SHOULD SERVICE THE EQUIPMENT. THE MANUFACTURER WILL NOT BE RESPONSIBLE FOR ANY INJURY OR PROPERTY DAMAGE ARISING FROM IMPROPER SERVICE OR SERVICE PROCEDURES. IF YOU SERVICE THIS UNIT, YOU ASSUME RESPONSIBILITY FOR ANY INJURY OR PROPERTY DAMAGE WHICH MAY RESULT. IN ADDITION, IN JURISDICTIONS THAT REQUIRE ONE OR MORE LICENSES TO SERVICE THE EQUIPMENT SPECIFIED IN THIS MANUAL, ONLY LICENSED PERSONNEL SHOULD SERVICE THE EQUIPMENT. IMPROPER INSTALLATION, ADJUSTMENT, SERVICING OR REPAIR OF THE EQUIPMENT SPECIFIED IN THIS MANUAL, OR ATTEMPTING TO INSTALL, ADJUST, SERVICE OR REPAIR THE EQUIPMENT SPECIFIED IN THIS MANUAL WITHOUT PROPER TRAINING MAY RESULT IN PRODUCT DAMAGE, PROPERTY DAMAGE, PERSONAL INJURY OR DEATH.**

**PROP 65 WARNING  
FOR CALIFORNIA CONSUMERS**

 **WARNING**

Cancer and Reproductive Harm -  
[www.P65Warnings.ca.gov](http://www.P65Warnings.ca.gov)

0140M00517-A

 **WARNING**

**HIGH VOLTAGE!**

**DISCONNECT ALL POWER BEFORE SERVICING OR INSTALLING THIS UNIT. MULTIPLE POWER SOURCES MAY BE PRESENT. FAILURE TO DO SO MAY CAUSE PROPERTY DAMAGE, PERSONAL INJURY OR DEATH.**



## ATTENTION INSTALLING PERSONNEL

As a professional installer, you have an obligation to know the product better than the customer. This includes all safety precautions and related items.

Prior to actual installation, thoroughly familiarize yourself with this Instruction Manual. Pay special attention to all safety warnings. Often during installation or repair, it is possible to place yourself in a position which is more hazardous than when the unit is in operation.

Remember, it is your responsibility to install the product safely and to know it well enough to be able to instruct a customer in its safe use.

Safety is a matter of common sense...a matter of thinking before acting. Most dealers have a list of specific good safety practices...follow them.

The precautions listed in this Installation Manual are intended as supplemental to existing practices. However, if there is a direct conflict between existing practices and the content of this manual, the precautions listed here take precedence.

|           | PART NO.   | DESCRIPTION                   | QTY |
|-----------|------------|-------------------------------|-----|
| TTBCKHE01 | 0163L00152 | ½" Gas Adapter Fitting        | 1   |
|           | 0163L00154 | ½" Elec. Bulkhead Connector   | 2   |
|           | 0163L00196 | 1-½" Elec. Bulkhead Connector | 1   |
|           | 0121L02301 | Bracket Thru the Base         | 1   |
|           | B1910010   | Gas Line Grommet              | 1   |
|           | B1393561   | #10 x ½" Self-Drilling Screw  | 2   |
| TTBCKHE02 | 0163L00154 | ½" Elec. Bulkhead Connector   | 2   |
|           | 0163L00197 | 2" Elec. Bulkhead Connector   | 1   |
|           | 0121L02876 | Bracket Thru the Base         | 1   |
|           | B1393561   | #10 x ½" Self-Drilling Screw  | 2   |

Table 1 - Thru-the-Bottom Connection Parts

## DESCRIPTION

Thru-the-bottom service connection kits reduce the number of penetrations in the roof. They provide a watertight way to connect power wiring, control wiring, and the gas supply line through the base of the unit in the condenser section. External connectors must be weatherproof. All holes in the unit base must be sealed (including those around conduit nuts) to prevent water leakage into the building.

When the thru-the-bottom power accessory package is purchased, one of two bulkhead connector sets may be selected for power connections (depending on unit size). Each set contains 3 to 4 bulkhead connectors, depending on unit tonnage (Table 1). The ½ in. diameter bulkhead connectors enable the low-voltage control, and optional convenience outlet, wires to pass through the basepan. The 1-½ or 2 in. bulkhead connector allows the high voltage power wires to pass through the basepan. The ½ in. gas adapter fitting allows the gas line to come up through the basepan (3 to 6 ton units only).



## INSTALLATION

Disconnect all power before installing this kit.

Make sure the roof curb is installed and leveled in accordance with the instructions provided with kit.

The thru-the-bottom service connection accessory package must be used with a light commercial unit that has embossment in the outdoor section of the basepan. This opening allows high-voltage power, control wiring, and gas connections to pass through the roof curb and into the basepan of the rooftop unit.

## PREPARING THE UNIT FOR INSTALLATION

The accessory thru-the-bottom power or gas connection package fittings must be installed before the unit is set on the roof curb. Gas and electrical connections to the unit must be field installed after the unit is installed on the roof curb.

## INSTALL THE UNIT ON THE ROOF CURB

Follow all appropriate instructions required to install the rooftop unit on the roof curb. See installation instructions provided with the roof curb and unit for more specific information.

## CREATING HOLES IN THE RAISED SURFACE OF THE UNIT BASEPAN

Each unit is equipped with a raised (embossed) surface in the basepan. This area is located in the compressor compartment (outdoor section) of the unit (see Figure 1). The embossment of the basepan contains three (3) dimples for locating cutouts. Holes can be field cut in the embossed surface so that high voltage power, control wires, and gas connection can be routed through the roof curb basepan

Drill holes for gas and power at the 3 dimples shown below.

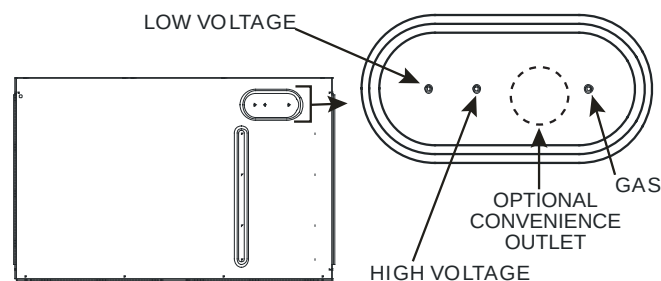


Fig. 1 - 3 - 6 Ton Basepan

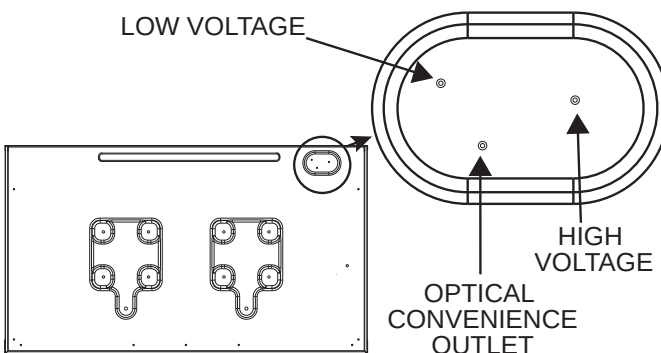


Fig. 2 - 7.5 - 12.5 Ton Basepan

Use a chassis punch or hole saw to create holes in the embossed surface of the basepan per Table 2.

| CONNECTION SIZE | USE          | REQ. HOLE SIZE |
|-----------------|--------------|----------------|
| 1/2"            | Low Voltage  | 7/8"           |
| 1/2"            | Gas          | 1"             |
| 3/4"            | Gas          | 1-1/8"         |
| 1-1/2"          | High Voltage | 2"             |
| 2"              | High Voltage | 2-1/2"         |

Table 2 - Bottom Power Chart

## MOUNTING ELECTRIC BULKHEAD CONNECTORS

Once the holes have been cut place electrical bulkhead connectors in basepan. Place gasket and lock ring end of connector beneath the basepan and mount the other end above the basepan.

When the electrical connectors are in place, field supplied and installed liquid tight conduit connectors and conduit may be attached to the connectors. Pull correctly rated high voltage and low voltage wires through appropriate conduits and connect to units per roof top units installation instructions.

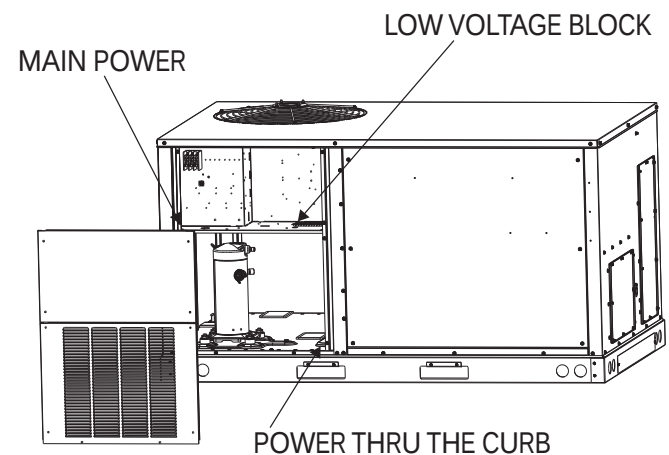


Fig. 3 - 3-6 Ton Power Connections

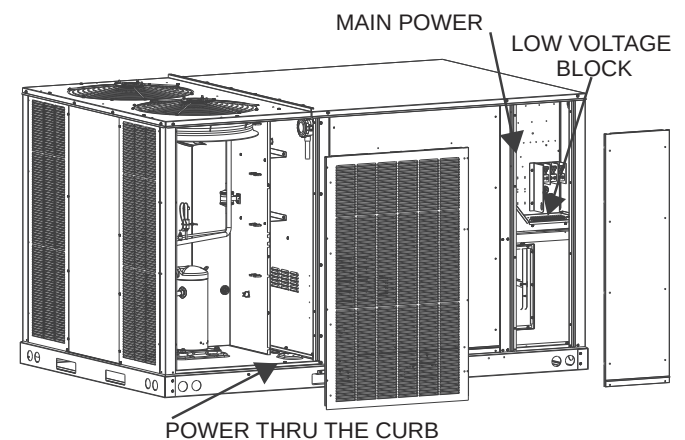


Fig. 4 - 7.5 - 12.5 Ton Power Connections

## MAKING GAS CONNECTIONS

### (3 TO 6 TON UNITS ONLY)

The thru-the-bottom gas connection fitting has male and female threads. The gas fitting also contains two neoprene washers and two locking washers. One neoprene and one locking washer are installed below the unit basepan with the second set being installed on the top side of the basepan before the fitting is tightened. Tighten all fittings. The male threads protrude above the basepan of the unit, while the female threads protrude below the basepan.

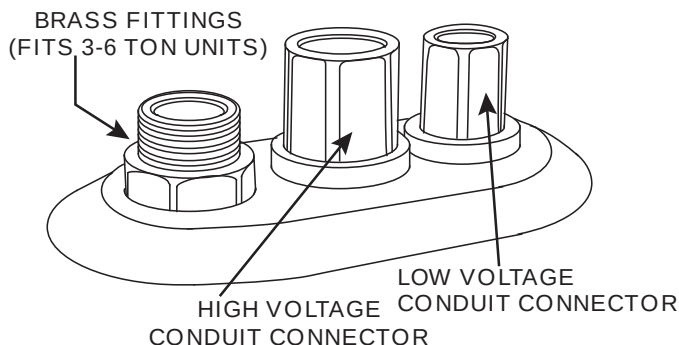


Fig. 5

Install a 5" black pipe nipple and an elbow on the through-the-basepan gas fitting (1/2-in. for 3-6 Ton kit).

If using 1/2-in. pipe, a minimum 12-in. nipple (field-supplied) is required to attach to the elbow and extends the gas piping through compressor access panel (if unit is so equipped). If using 3/4-in. pipe, a minimum 15-in. nipple (field-supplied) is required to extend the gas piping through compressor access panel (if unit is so equipped). Install the gas pipe support bracket to the inside of unit using provided screws. Remove the bottom right ten louvers from the right side of the louvered compressor access panel to allow gas line to exit the unit.



Fig. 6



Fig. 7



Fig. 8

## INSTALLING GAS PIPING

### (3 TO 6 TON UNITS ONLY)

The rating plate is stamped with the model number, type of gas and gas input rating. Make sure the unit is equipped to operate on the type of gas available. Conversion to propane (LP) gas is permitted with the use of the factory authorized conversion kit (see the unit Technical Manual for the appropriate kit). For High Altitude derates, refer to the latest edition of the National Fuel Gas Code NFPA 54/ANSI Z223.1.

A manual gas shut off valve must be field installed external to the roof top unit. In addition, a drip leg must be installed near the inlet connection. A ground joint union connection is required between the external shut off valve and the unit connection to the gas valve to permit removal of the burner assembly for servicing.

**NOTE:** Inlet gas pressures must not exceed the maximum value shown in Table 3 below.

| INLET GAS PRESSURE |                                  |
|--------------------|----------------------------------|
| NATURAL            | Min. 5.0" W.C., Max. 10.0" W.C.  |
| PROPANE            | Min. 11.0" W.C., Max. 14.0" W.C. |

Table 3 - Inlet Gas Pressure

**CUSTOMER FEEDBACK**

Daikin is very interested in all product comments.

Please fill out the feedback form on the following link:

<https://daikincomfort.com/contact-us>

You can also scan the QR code on the right to be directed to the feedback page.



NOTE: SPECIFICATIONS AND PERFORMANCE DATA LISTED HEREIN ARE SUBJECT TO CHANGE WITHOUT NOTICE

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[www.daikincomfort.com](http://www.daikincomfort.com)

**DAIKIN****COMMERCIAL WARRANTY****Models: DRC, DRG, DRH, DHC, DHG, DHH**Who Is Providing The Warranty?

This warranty is provided to you by Daikin Comfort Technologies Manufacturing, L.P. ("Daikin"), which warrants all parts of this heating or air conditioning unit, as described below.

To What Type Of Installations Does This Warranty Apply?

This warranty applies to heating and air conditioning units installed in buildings other than residences.

What Units Does This Warranty Not Cover?

This warranty does not apply to:

- Units that are ordered over the Internet, by telephone, or by other electronic means unless the unit is installed by a dealer adhering to all applicable federal, state, and local codes, policies, and licensing requirements.
- Units that are installed outside the United States, its territories, or Canada.
- Units that are operated in incomplete structures.
- Units that are installed in residential buildings.

What Problems Does This Warranty Cover?

This warranty covers defects in materials and workmanship that appear under normal use and maintenance.

Other Warranties

**THIS WARRANTY IS PROVIDED IN LIEU OF ANY OTHER WARRANTIES, WHETHER BY DAIKIN OR ANY OF ITS AFFILIATES, EXPRESS OR IMPLIED, INCLUDING ANY IMPLIED WARRANTY OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE.**

What Problems Does This Warranty Not Cover?

Daikin is not responsible for:

- Damage or repairs required as a consequence of faulty installation or application.
- Damage as a result of floods, fires, winds, lightning, natural disasters, accidents, corrosive atmosphere, or other conditions beyond Daikin's control.
- Damage or the need for repairs arising from the use of components or accessories not compatible with this unit.
- Normal maintenance as described in the installation and operating manual, such as cleaning of the coils, filter cleaning and/or replacement, and lubrication.
- Parts or accessories not supplied or designated by the manufacturer.

- Damage or the need for repairs resulting from any improper use, maintenance, operation, or servicing.
- Damage or failure of the unit to start due to interruption, surge, or fluctuation in electrical service or inadequate electrical service.
- Any damage, or the need for any repairs, caused by frozen or broken water pipes, water damage, moisture intrusion, mold or other biological growth. Changes in the appearance of the unit that do not affect its performance.
- Replacement of fuses and replacement or resetting of circuit breakers.
- Damage or the need for repairs resulting from the use of unapproved refrigerant types or used or recycled refrigerant.

When Does Warranty Coverage Begin?

Warranty coverage begins on the "installation date." The installation date is one of two dates:

- (1) The installation date is the date that the unit is originally installed.
- (2) If the date the unit is originally installed cannot be verified, the installation date is three months after the manufacture date. The first four digits of the serial number (YYMM) on the unit indicate the manufacture date. For example, a serial number beginning with "2409" indicates the unit was manufactured in September 2024.

How Long Does Warranty Coverage Last?

The warranty lasts for a period of up to (1) 20 YEARS after the installation date for the heat exchanger (only applicable to units containing a heat exchanger) and (2) 5 YEARS after the installation date on all other parts.

This warranty period does not continue after the unit is removed from the location where it was originally installed.

The replacement of a part under this warranty does not extend the warranty period. In other words, Daikin warrants a replacement part only for the period remaining under the applicable warranty on the original part.

**www.daikincomfort.com**

For further information about this warranty,  
contact Homeowner Support by mail to  
19001 Kermier Road, Waller, Texas 77484

Part No. PWDLCPPHED  
09/2024

**DAIKIN****COMMERCIAL WARRANTY****Models: DRC, DRG, DRH, DHC, DHG, DHH**What Will Daikin Do To Correct Problems?

Daikin will furnish a replacement part, without charge for the part only, to replace any part that is found to be defective due to workmanship or materials under normal use and maintenance during the warranty period. Furnishing of the replacement part is Daikin's only responsibility under this warranty.

**THE REMEDIES DESCRIBED IN THIS SECTION ARE DAIKIN'S ONLY RESPONSIBILITIES, AND THE OWNER'S ONLY REMEDIES, FOR ANY BREACH OF ANY WARRANTY.**

What Won't Daikin Do To Correct Problems?

Daikin will not pay for:

- Labor, freight, or any other cost associated with the service, repair, or operation of the unit, the deinstallation of any defective part, or the installation of any replacement part.
- Electricity or fuel costs, or increases in electricity or fuel costs, for any reason, including additional or unusual use of supplemental electric heat.
- Lodging or transportation charges.
- Refrigerant.

**WHETHER ANY CLAIM IS BASED ON NEGLIGENCE OR OTHER TORT, BREACH OF WARRANTY OR OTHER BREACH OF CONTRACT, OR ANY OTHER THEORY, NEITHER DAIKIN NOR ANY OF ITS AFFILIATES SHALL IN ANY EVENT BE LIABLE FOR INCIDENTAL OR CONSEQUENTIAL DAMAGES, INCLUDING BUT NOT LIMITED TO LOST PROFITS, LOSS OF USE OF A UNIT, EXTRA UTILITY EXPENSES, OR DAMAGES TO PROPERTY.**

How Can The Owner Receive Warranty Service?

If there is a problem with the unit, contact a licensed contractor.

To receive a replacement part, a licensed contractor must bring the defective part to a Daikin heating and air conditioning products distributor.

For more information about the warranty, write to Homeowner Support, 19001 Kermier Road, Waller, Texas 77484.

Where Can Any Legal Remedies Be Pursued?

ARBITRATION CLAUSE. IMPORTANT. PLEASE REVIEW THIS ARBITRATION CLAUSE. IT AFFECTS YOUR LEGAL RIGHTS.

1. **Parties:** This arbitration clause affects your rights against Daikin Comfort Technologies North America, Inc. and any of its subsidiaries, including but not limited to Daikin Comfort Technologies Manufacturing, L.P., and its or their employees or agents, successors, or assigns, all of whom together are referred to below as "we" or "us" for ease of reference.

2. **ARBITRATION REQUIREMENT: EXCEPT AS STATED BELOW, ANY DISPUTE BETWEEN YOU AND ANY OF US SHALL BE DECIDED BY NEUTRAL, BINDING ARBITRATION RATHER THAN IN COURT OR BY JURY TRIAL.** "Dispute" will be given the broadest possible meaning allowable by law. It includes any dispute, claim, or controversy arising from or relating to your purchase of this heating or air conditioning unit, any warranty upon the unit, or the unit's condition. It also includes determination of the scope or applicability of this Arbitration Clause. The arbitration requirement applies to claims in contract and tort, pursuant to statute, or otherwise.
3. **CLASS-ARBITRATION WAIVER: ARBITRATION IS HANDLED ON AN INDIVIDUAL BASIS. IF A DISPUTE IS ARBITRATED, YOU AND WE EXPRESSLY WAIVE ANY RIGHT TO PARTICIPATE AS A CLASS REPRESENTATIVE OR CLASS MEMBER ON ANY CLASS CLAIM YOU MAY HAVE AGAINST US OR WE AGAINST YOU, OR AS A PRIVATE ATTORNEY GENERAL OR IN ANY OTHER REPRESENTATIVE CAPACITY, TO THE MAXIMUM EXTENT PERMITTED BY LAW. YOU AND WE ALSO WAIVE ANY RIGHT TO PARTICIPATE IN ANY CLASS OR REPRESENTATIVE ACTION IN ANY FORM, INCLUDING ANY CLASS ARBITRATION, OR ANY CONSOLIDATION OF INDIVIDUAL ARBITRATIONS.**
4. **Discovery and Other Rights:** Discovery and rights to appeal in arbitration are generally more limited than in a lawsuit. This applies to both you and us. Other rights that you or we would have in court may not be available in arbitration. Please read this Arbitration Clause and consult the rules of the arbitration organizations listed below for more information.
5. **SMALL CLAIMS COURT OPTION: YOU MAY CHOOSE TO LITIGATE ANY DISPUTE BETWEEN YOU AND ANY OF US IN SMALL CLAIMS COURT, RATHER THAN IN ARBITRATION, IF THE DISPUTE MEETS ALL REQUIREMENTS TO BE HEARD IN SMALL CLAIMS COURT.**
6. **Governing Law:** For residents of the United States, the procedures and effect of the arbitration will be governed by the Federal Arbitration Act (9 U.S.C. § 1 et seq.) rather than by state law concerning arbitration. For residents of Canada, the procedures and effect of the arbitration will be governed by the applicable arbitration law of the province in which you purchased your unit. The law governing your substantive warranty rights and other claims will be the law of the state or province in which your unit was installed. Any court having jurisdiction may enter judgment on the arbitration award.

Part No. PWDLCPPHED  
09/2024

**DAIKIN****COMMERCIAL WARRANTY****Models: DRC, DRG, DRH, DHC, DHG, DHH**

7. *Rules of the Arbitration:* If the amount in controversy is less than \$250,000, the arbitration will be decided by a single arbitrator. If the amount in controversy is greater than or equal to \$250,000, the arbitration will be decided by a panel of three arbitrators. The arbitrator(s) will be chosen pursuant to the rules of the administering arbitration organization. United States residents may choose JAMS (1920 Main Street, Ste. 300, Irvine, CA 92614, [www.jamsadr.com](http://www.jamsadr.com)) or, subject to our approval, any other arbitration organization. In addition, Canadian residents may choose the ADR Institute of Canada (234 Eglinton Ave. East, Suite 405, Toronto, Ontario, M4P 1K5, [www.amic.org](http://www.amic.org)). These organizations' rules can be obtained by contacting the organization or visiting its website. If the chosen arbitration organization's rules conflict with this Arbitration Clause, the provisions of this Arbitration Clause control. The award of the arbitrator(s) shall be final and binding on all parties.
8. *Location of the Arbitration Hearing:* Unless applicable law provides otherwise, the arbitration hearing for United States residents will be conducted in the federal judicial district in which you reside (in your hometown area) or, for Canadian residents, in the province in which you reside, and if you choose, will be in person.
9. *Costs of the Arbitration:* Each party is responsible for its own attorney, expert, and other costs and fees unless applicable law requires otherwise. Each party is also responsible for one-half of any cost and fees charged by the arbitration organization and arbitrator(s) to administer the arbitration to the maximum extent permitted by law or rule. Where permissible by law, the prevailing party may be required to reimburse the other party for the costs and fees of the arbitration organization and arbitrator(s) in whole or in part by decision of the arbitrator(s) at the discretion of the arbitrator(s).
10. *Survival and Enforceability of this Arbitration Clause:* This Arbitration Clause shall survive the expiration or termination, or any transfer, of the warranty on your unit. If any part of this Arbitration Clause, except waivers of class-action rights, is found to be unenforceable for any reason, the remainder of this clause and the warranty shall remain enforceable. If, in a case in which class-action allegations have been made, the waiver of class-action rights under this warranty is found to be unenforceable with respect to any part of the dispute, the parts of the dispute as to which the waiver of class-action rights have been found unenforceable will be severed and will proceed in court without reference or application of this Arbitration Clause. Any remaining parts will proceed in arbitration.

Part No. PWDLCPPHED  
09/2024